



Phase change materials: classification, use, phase transitions, and heat transfer enhancement techniques: a comprehensive review

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Abstract

Currently, there is great interest in producing thermal energy (heat) from renewable sources and storing this energy in a suitable system. The use of a latent heat storage (LHS) system using a phase change material (PCM) is a very efficient storage means (medium) and offers the advantages of high volumetric energy storage capacity and the quasi-isothermal nature of the storage process. In recent years, phase change materials (PCMs) have become an interesting research area due to their advantages especially in thermal energy storage (TES). Indeed, there are a large number of PCMs that melt and solidify over a wide temperature range, making them interesting thermal energy storage media in several applications. In the literature, research on PCMs and their associated applications has attracted and still attracts great interest from various researchers and scientists. Most of the research studies on phase change materials (PCMs) have been generally devoted to the development of PCM-based energy storage technologies, the promotion of PCM-based renewable energy sources, and the encouragement of sustainable/profitable (economic) use of PCM-based energy. In order to get an overview of current progress and trends, to highlight research and to identify gaps, from the literature reviews undertaken on this research topic, it is useful to review the major research studies conducted in this field. Our analysis showed that the literature lacks many comprehensive analyses and studies on the applications of PCMs, the phase transition processes (melting and solidification) of PCMs and the factors that influence these transitions, and in particular the calculation models of the thermal performance parameters of a PCM performing a phase transition and the thermal performance parameters of a PCM-based TES system (referred to as LHS unit). To address these questions, we have presented in this review article a detailed overview of the literature on (a) relevant practical applications of PCMs, (b) characteristics and performances of phase transition processes, (c) major factors influencing PCM transition processes such as geometric design of the PCM tank and its orientation, imposed boundary and operating conditions, thermophysical properties of the material (PCM), and (d) models for calculating thermal performance parameters for a PCM performing a phase transition and for an LHS unit. In addition, several techniques aimed at improving heat transfer in PCMs have been introduced and discussed. The findings indicate that there are three types of PCMs: eutectic, inorganic, and organic. Numerous other industries also use PCMs, such as solar energy (including thermal energy storage through the use of photovoltaic and latent heat systems); buildings; HVAC systems; textiles; the biomedical, food, and agricultural industries; the automotive sector; and desalination. Besides PCMs classification and use, it was found that during phase transitions of PCMs heat transfer is dominated by conduction and natural convection. During melting, conduction heat transfer is dominant in the early stages, and as the PCM melts, natural convection dominates. Unlike melting, solidification is dominated by conductive heat transfer. On the other hand, boundary conditions, material properties, and enclosure configuration and orientation all found having an impact on melting and solidification. In this context, by increasing, for example, thermal conductivity, viscosity, wall-imposed temperature, and PCM initial temperature, as well as by decreasing PCM latent heat of melting, PCM melting point, and PCM system orientation, the melting process rate increases. However, by increasing thermal conductivity, viscosity, melting point, and PCM system orientation, as well as by lowering the latent heat of melting, the initial PCM temperature, and the imposed wall temperature, the solidification process rate increases. Lastly, introducing external fields and adding high thermal conductivity additives like fins, metal foam, and nanoparticles can greatly increase the rate at which PCM melts and solidifies.

Keywords Phase change material · Latent heat storage · Melting · Solidification · Heat transfer performance enhancement techniques

Abbreviations

TES	Thermal energy storage
SHS	Sensible heat storage
TCS	Thermochemical heat storage
LHS	Latent heat storage
LHTES	Latent heat thermal energy storage
PS-TES	Pear-shaped thermal energy storage
PCM	Phase change material
Ne-PCM	Nano-enhanced phase change material (or PCM enhanced by nanoparticle dispersions)
HNe-PCM	Hybrid nano-enhanced phase change material
SSPCM	Shape-stabilized phase change material
PCMS	Phase change material suspensions
MPCMS	Microencapsulated phase change material slurry
HTF	Heat transfer fluid
HVAC	Heat, ventilation, and air conditioning
HX	Heat exchanger
THX	Tubular heat exchanger
DTHX	Double-tube heat exchanger
TTHX	Triplex tube heat exchanger
QTHX	Quadruple-tube heat exchanger
HP	Heat pipe
3D-OHPs	Three-dimensional oscillating heat pipes
HS	Heat sink
NTUs	Number of heat transfer units
CA-LA	Capric acid—lauric acid ((belongs to the category of fatty acids))
CA-PA	Capric acid—Palmitic acid (belongs to the category of fatty acids)
PEG	Polyethylene glycol (PCM)
SWCNTs	Single-walled carbon nanotubes
MWCNTs	Multi-walled carbon nanotubes
DM-CNTs	D-Mannitol—carbon nanotubes (i.e., nanocomposite phase change materials)
DM-MWCNTs	D-Mannitol—multi-walled carbon nanotubes
GNPs	Graphene nanoplatelets (e.g., dispersion of graphene nanoplatelets)
GNs	Graphite nanosheets (e.g., dispersion of graphite nanosheets)
PVP	Polyvinylpyrrolidone (a nanoparticle dispersant)
PMMA	Poly(methyl-methacrylate)

SSL	Sodium stearyl lactylate
HMPC	Hybrid model predictive controller

Fields and coordinates

u, v, w	Velocity field components of convective flow in PCM liquid phase, m s^{-1}
T	Temperature field, K
H	Enthalpy, J
χ	Concentration (composition) field, mass% (mass percent) or vol% (volume percent)
f_s	Solid fraction of the PCM (mass fraction of PCM solid)
f_l	Liquid fraction of the PCM (mass fraction of PCM liquid) ($f_l = 1 - f_s$)
ε_l	Volume fraction of PCM liquid (i.e., porosity)
g	Gravity, m s^{-2}
x, y, z	Cartesian coordinates of the PCM storage system (e.g., rectangular or cuboid domain), m^{-1}
r, θ, z	Cylindrical coordinates of the PCM storage system (e.g., cylinder or tubular domain), m^{-1}
t	Time, s

Thermophysical properties

H_L	Latent heat of phase change (specific enthalpy of phase change) of the PCM, J kg^{-1}
k	Thermal conductivity of the PCM (i.e., we distinguish k_s for solid and k_l for liquid PCM), $\text{W m}^{-1} \text{K}^{-1}$
C_p	Specific heat (heat capacity) of the PCM (we distinguish C_{ps} for solid and C_{pl} for liquid PCM), $\text{J kg}^{-1} \text{K}^{-1}$
α	Thermal diffusivity of the PCM (i.e., we distinguish α_s for solid and α_l for liquid PCM), $\text{m}^2 \text{s}^{-1}$
ρ	Density of the PCM (i.e., we distinguish ρ_s for solid and ρ_l for liquid PCM), kg m^{-3}
μ	Viscosity of the PCM liquid, Pa s^{-1}
ν	Kinematic viscosity of the PCM liquid, $\text{m}^2 \text{s}^{-1}$
β	Thermal expansion coefficient of the PCM liquid, K^{-1}

Performance parameters of a PCM (domain) undertaking a phase transition

T_0	Initial temperature of the PCM at time ($t = 0s$) (can also be denoted by T_{in} ; at $t = 0s$, we have $T_0 = T_{in}$), K
T_w	Temperature imposed on the wall of the PCM storage domain (PCM tank), K
T_c	Temperature imposed on the cold wall of the PCM storage domain, K
T_h	Temperature imposed on the heated wall of the PCM storage domain, K
T_i	Initial temperature of the PCM (before the phase change) (if melting $T_i = T_c$), K
T_f	Final temperature of the PCM (after the phase change) (if melting $T_f = T_h$), K
T_m	Phase change temperature of the PCM (if melting it represents the melting point), K
m	Mass of PCM substance in the storage domain (storage tank), kg
m_{s-l}	Mass of the molten PCM in the domain ($m_{s-l} = mf_l$), kg
H, L	Height (width) and length of rectangular PCM storage system (e.g., size of storage domain), m^{-1}
V_{tank}	Volume of PCM storage domain (can be also designated by V_{pcm}), m^{-3}
R, D, L	Radius, diameter, and height of cylindrical-shaped PCM storage system (cylindrical shell), m^{-1}
U_{ref}	Reference velocity of PCM melt flow, $m s^{-1}$
t_m	The total duration of a complete cycle (storage time) (if melting it represents melting time), s
$\dot{q}_m$	Heat transfer rate (can also be denoted by $\dot{q}_{pcm}$; if melting it represents the melting rate), W (or $J s^{-1}$)
$E_{S,i}$	Sensible heat stored by PCM before phase change, J
$E_{S,f}$	Sensible heat stored by PCM after phase change, J
E_S	Total sensible heat stored by PCM (before and after phase change), J
$E_{L,m}$	Latent heat stored by PCM during the phase change, J
E	Total energy (heat) stored by PCM (or total storage energy amount), J
E_V	Volumetric heat storage capacity (or volumetric energy storage density) of the PCM, $W m^{-3}$ (or $MWh m^{-3}$)

Performance parameters of an LHS unit

$C_{p,HTF}$	Specific heat of the HTF fluid passing through the storage system (through the inner channel), $J kg^{-1} K^{-1}$
$T_{out,HTF}$	Temperature of the hot HTF fluid at the outlet of the storage system channel “desired” (prepared) for the use (e.g., domestic heating), K
$T_{in,HTF}$	Temperature of the cold HTF fluid, which arrives at the inlet of the storage system channel (supply temperature of the storage system with the HTF fluid), K
m_{HTF}	“Required” mass of the HTF passing through the storage system channel, kg
$\dot{m}_{HTF}$	Required mass flow rate of the HTF passing through the storage system channel, $kg s^{-1}$
$T_{max,HTF}$	Maximum temperature of the HTF fluid flow in the storage system (sensible heat storage temperature), K
$T_{av,HTF}$	Average temperature of the HTF flow in the channel (approximated by $T_{av,HTF} = (T_{max,HTF} + T_{in,HTF})/2$), K
T_i	Initial temperature of the storage medium (PCM) (i.e., if melting $T_i = T_c = T_{in,HTF}$), K
T_f	Final temperature of the storage medium (PCM) (i.e., if melting $T_f = T_h = T_{out,HTF}$), K
$\dot{q}_{HTF}$	Heat transfer rate (energy stored per unit of time) in the LHS unit, W
$E_{S,HTF}$	Sensible energy stored by the HTF, J
$E_{S,i LHS}$	Sensible heat stored by PCM (in LHS unit) before phase change due to temperature variation $\Delta T_{i \rightarrow m} = (T_m - T_i)$, J
$E_{S,f LHS}$	Sensible heat stored by PCM (in LHS unit) after phase change due to temperature variation $\Delta T_{m \rightarrow f} = (T_f - T_m)$, J
$E_{S,LHS}$	Total sensible heat stored by PCM (in the LHS unit) before and after phase change, J
$E_{L,m LHS}$	Latent heat stored by PCM (in the LHS unit) during the phase change, J
E_{LHS}	Energy storage capacity of (total energy stored in) the LHS unit, J
$t_{HTF,Use}$	Time for storage of hot HTF in PCM heat storage tank “prepared” for use, s
$m_{HTF,Use}$	Mass of hot HTF in PCM heat storage tank “prepared” for use, kg

Introduction

Over the coming decades, continued rapid development of the global economy and population growth will follow strong demand for energy supplies. According to British Petroleum, by 2035, the world's population will reach 8.7 billion and global energy consumption will increase by 37% [1].

Energy (heat) produced from fossil fuels has always satisfied and responded to the needs of humanity since the appearance of these resources. On the other hand, the limited availability (no abundance) or even the exhaustion of these resources and the consequences (on health and the environment) resulting from their use such as the emission of harmful gases and global warming have raised several challenges and concerns. These major issues have led to the development and adoption of renewable energy sources due to their abundance and availability. According to the International Energy Agency's prediction, by 2030, the proportion of energy production that comes from renewable sources will rise from about 29% in 2021 to over 60% [2].

However, due to the intermittency and instability of renewable energy control, thermal energy storage (TES) can overcome the disadvantage of renewable energy systems by storing the energy generated during off-peak hours of consumption (high consumption) and returning (recovering/releasing) this energy during the period of low consumption. Thermal energy storage (TES), which uses a storage medium (a continuum medium), can occur in all three modes: sensible heat storage (SHS), latent heat storage (LHS), and thermochemical heat storage (TCS) (see Fig. 1). Sensible heat storage [3, 4] consists of adding energy to a material (the storage medium) to increase its temperature without changing phase. Latent heat storage [5, 6] relies on the release or absorption (storage) of heat (called latent heat) when the phase of the storage material changes from liquid to solid or vice versa, respectively. Thermochemical storage (TCS) [7–9] occurs at the molecular level after a chemical band-breaking or dissociation reaction, thereby releasing energy and recovering it in a reversible reaction. Among the

previous storage techniques, the storage of latent heat that occurs in phase change materials (PCMs) is considered a promising option, because these materials enable volumetric heat storage at high capacity (density), and are characterized by their ability to charge energy (during melting) and discharge energy (during solidification) under quasi-isothermal operating conditions [10].

A phase change material (PCM) is a substance made up of molecules that is primarily used for storing thermal energy. The principle behind its function is straightforward: when the temperature rises, the material undergoes a phase change from solid to liquid (melting) and absorbs energy during this process. Thus, melting (energy charging process) is an endothermic transformation. Similarly, when the temperature decreases, the PCM in the liquid phase solidifies and releases (desorbs) energy and the solidification (energy-discharging process) is an exothermic reaction. These exceptional characteristics of PCMs make their use as a thermal energy storage medium in engineering very broad and promising. Indeed, energy storage using phase change materials is today used in numerous practical applications, for example: in latent heat storage systems (called LHS units), in construction materials (e.g., embedding PCM in brick) used in buildings, gymnasiums, etc., in *cooling* devices for electronic apparatus, in HVAC systems, in the textile industry, in the biomedical field, etc. It should be noted here that an LHS unit is a storage system consisting essentially of a PCM tank comprising one (or more) internal channels. The internal (inner) channel is traversed by the flow of heat transfer fluid “HTF,” which ensures either the release of latent heat from the LHS unit (in the event of solidification of PCM in the tank) or the storage of latent heat in the LHS unit (in the event of PCM melting in the tank). More details on this context are given below.

Nevertheless, despite the aforementioned advantages of PCMs (i.e., high volumetric heat storage capacity and quasi-isothermal operating conditions), it is imperative to note that PCMs have drawbacks. Typically, a PCM is characterized by poor thermophysical properties, including its thermal conductivity and thermal diffusivity. The low thermal conductivity and thermal diffusivity of PCM reduces the *heat transfer rate* during melting and/or solidification (i.e., melting rate and/or solidification rate), which increases the overall *duration* (time) of melting and/or solidification cycles; which in turn can cause large temperature differences in the PCM, causing the LHS storage system to overheat and potentially cause hardware failure [11]. On the other hand, due to the low heat capacity of PCM, LHS systems see their productivity (in terms of *energy stored*) weakened. Therefore, improvements in thermal conductivity, thermal diffusivity, heat capacity, and other properties/characteristics of PCM (among other means discussed below) are essential to increase the heat transfer rate and energy stored, during

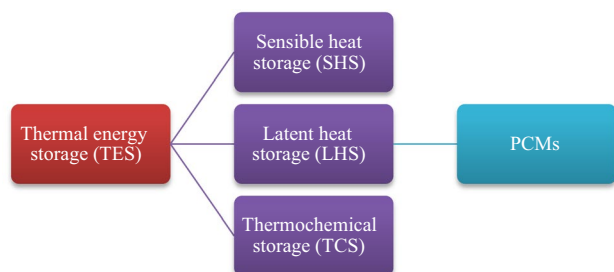


Fig. 1 Overview of different types of thermal energy storage

the phase change undertaken by the PCM, and to reduce the time of this transition; which thus leads to increasingly efficient PCM-based storage systems [12]. The heat transfer rate, energy stored, and time of the transition process represent the main thermal performances of a PCM-based storage system.

The use of PCM types and heat transfer enhancement techniques for LTE devices has been extensively reviewed and published. The unique thermal characteristics of PCMs, their classification and working principles, and the advances in PCM applications in various industries are all summarized in detail by the study of Reddy et al. [13]; which also highlights the enormous potential of PCMs to increase energy efficiency, reduce carbon emissions, and support sustainable development. Salman et al. [14] explored the integration of phase change materials (PCMs) with building materials and reviewed various experimental and numerical methods to evaluate the thermal performance of these PCM-containing materials. The three types of heat transfer interfaces (HTIs) between PCM and heat transfer fluid (HTF): flat surface, extended surface, and direct contact were used by Tian et al. [15] to calculate and evaluate the overall convective heat transfer coefficients on the PCM side. The comparison results highlight the importance of increased accessibility between PCM and HTF and demonstrate that expandable and diversified HTIs are essential to improve heat transfer performance. Various techniques employed by researchers to improve the characteristics and thermal conductivity of PCMs are covered by Hussein et al. [16]. The authors also discussed the factors that influence the performance of PCM as TES, including PCM type, encapsulation type, TES geometry, and phase change temperature that should be selected based on the application. In their review of recent developments regarding the use of various types of metallic fins in LHTES, Low et al. [17] summarized the experimental and numerical research for charge and discharge cycles in LHTES, with emphasis on the effects of each fin parameter (length, angle, location, pitch, material, thickness, perforation, etc.). In addition to providing insight into the logical design and controlled preparation of PCM catalysts composites, Wang et al. [18] highlighted new developments in phase change materials to improve the catalytic process.

It should be emphasized, however, that the literature does not contain much in-depth analysis of PCM applications, heat transfer mechanisms, or how parameters affect the phase change process. This review discusses the classification and main use of PCMs in various thermal engineering systems encountered in practice. Indeed, there are different categories of phase change materials which have been classified according to various and varied classification criteria, the main criteria of which we attempt to clarify here. Also, to highlight the main transformations (melting and solidification) undergone by a phase change material, this

review pays particular attention to the main characteristics of the melting and solidification phase transition processes occurring in “a system of latent heat storage” itself (LHS unit). Indeed, an in-depth analysis of the main parameters that affect these processes, including the geometric shape of the storage system (PCM tank), the conditions and flow characteristics of the heat transfer fluid (mainly the flow rate of HTF and the “initial” temperature of HTF at the inlet of the inner channel of the LHS unit), the operating conditions of the storage unit, etc. were provided. This review also highlights the thermal performance of PCM-based storage systems and discusses various techniques from the literature to enhance this performance. These performance improvement techniques are essentially based on the one hand on the association of metal foams with the PCM or on the inclusion (insertion) of dispersion of nanoparticles into the PCM; or on the other hand on the incorporation of fins in the LHS system or the application of external fields to the system.

To address the growing need for efficient and innovative energy solutions, this study aims to not only fill the gaps in existing knowledge but also provide a comprehensive exploration of the applications and phase transition processes of PCMs and reveal recent advances to enhance performance or outcomes in the thermal performance of storage systems. This study will facilitate, for example, the selection of efficient LHS units, composed of reliable phase change materials confined in compatible and optimal tanks, presenting efficient geometric configurations.

Classification and use of PCMs

The nature and use of the phase change material are the most important standards (criteria) that must be considered in the design of the PCM-based thermal storage system.

Classification of phase change materials

Phase change materials can be classified into solid–solid, solid–liquid, solid–gas, and liquid–gas materials, as shown in Fig. 2 [19]. Solid–liquid phase change materials (PCMs), including organic, inorganic, and eutectic types, are the most suitable for latent heat storage (LHS) applications. A comparison of these different material types is presented in Table 1. Due to this variety of material categories, the selection of a PCM depends on several criteria, which are presented in Table 2.

Organic phase change materials encompass paraffinic and non-paraffinic substances, including fatty acids, alcohols, and glycols [20]. These organic PCMs have attracted attention due to their increased latent heat storage capacity, suitable phase transition temperature, and physical and chemical stability. However, they have some drawbacks that hinder

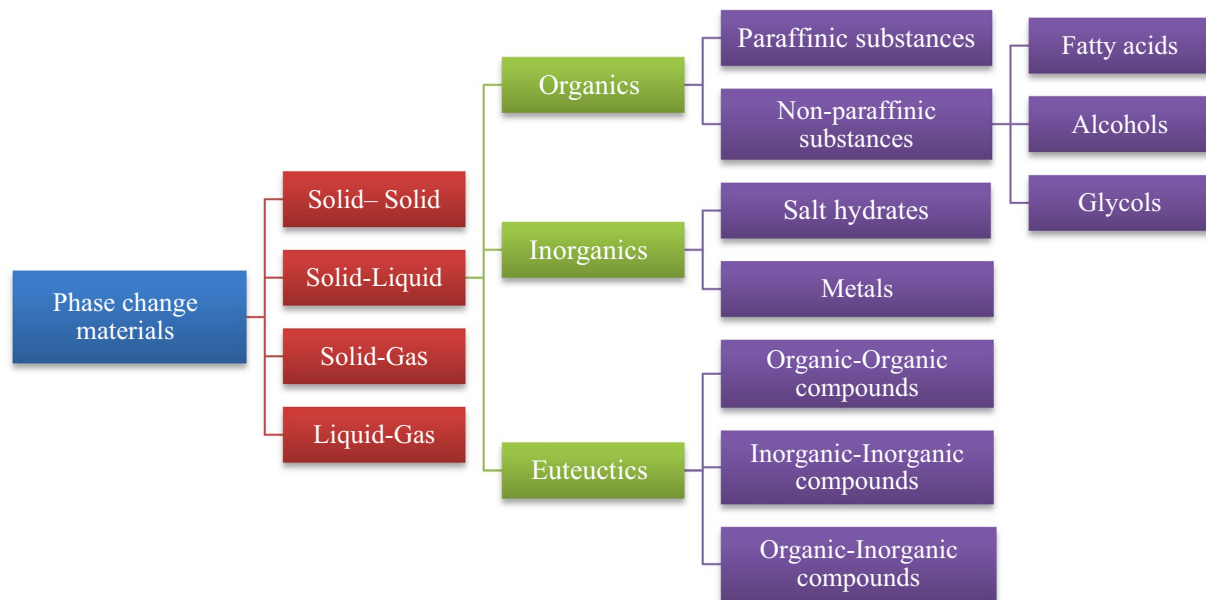


Fig. 2 Classification of phase change materials according to their physical state and nature [19]

Table 1 Advantages and disadvantages of different categories (classes) of phase change materials [24]

Classification	Advantages	Disadvantages
Organic PCMs	Versatility over a wide temperature range High latent heat of phase change Absence of subcooling Physical/chemical stability and recyclability High compatibility with various materials	Low thermal conductivity Significant volume change Low volumetric heat storage density Susceptibility to flammability
Inorganic PCMs	High melting point Excellent thermal conductivity Low volume change High volumetric heat storage density (compared to organic PCMs) Low-cost access	Supercooling Corrosion
Eutectics PCMs	Distinct phase change temperature High volumetric heat storage capacity	Insufficient data available on thermophysical properties

Table 2 Criteria for selecting phase change materials [25]

Thermophysical properties	Ideal melting point range High value of latent heat of phase change excellent thermal conductivity Significant specific heat Good density Minimal volume fluctuations and low vapor pressure Consistent melting
Kinetic properties	Fast nucleation rate to avoid supercooling Rapid crystallization rate which leads to optimal heat release Fast melting rate which leads to optimal heat storage capacity
Chemical properties	Full reversibility of solidification and melting Chemical stability and resilience Integrity maintained through numerous solidification/melting cycles No-corrosive No-toxic, no-flammable, and no-explosive
Sustainability, profitability, and economic efficiency	Reasonable cost Immediately accessible and widely available

their practical use, including low thermal conductivity, and undergo large volume changes and release of liquid during their phase transitions [21].

Inorganic phase change materials are the fundamental components used in latent heat storage. They can be classified into two main groups: (1) hydrated salts, characterized by a combination of inorganic salts and water forming a crystalline solid (ice), and (2) metals, which have not been seriously considered taken into account in PCM-based LHS technology, mainly due to their mass-related disadvantages [22].

Eutectic materials are a distinct category of phase change materials resulting from the mixture of organic–organic or inorganic–inorganic or organic–inorganic compounds with specific atomic ratios (concentrations). These materials (called alloys) solidify (freeze) and melt as a mixture of crystals, providing unique thermal properties [23].

Use of phase change materials

Latent heat storage in an LHS unit

As shown in Fig. 1, an LHS storage unit is a “thermal energy storage” (TES) system that uses phase change material as the storage medium. Since the phase transitions of PCMs are characterized by latent heat storage (absorption or release), a PCM-based TES unit is also called: an LHS storage unit (consisting mainly of a PCM tank crossed by a flow of heat transfer fluid (HTF)). In practice, there are several geometric shapes adopted for the physical domain of the LHS storage unit, including, for example, (a) the double-tube LHS storage unit, shown in Fig. 3a, consists of two concentric horizontal elliptical pipes: the inner pipe (tube) in which the HTF flows and the outer pipe (the annular space) is filled with PCM (i.e., the PCM tank) [26], (b) the storage unit in Fig. 3b which uses a modified multi-tube heat exchanger (THX) filled with PCM [27], and (c) the shell-and-tube LHS unit, which consists of a shell-and-tube heat exchanger as shown in Fig. 3c (this last storage unit can be used, for example, in the solar water heating system [28]). Another representative drawing of a typical LHS storage unit used, for example, in cooling/heating of a house is shown in Fig. 3d. In the literature, several other geometric shapes of LHS units have been used, as shown in Table 3.

Other applications of phase change materials as TES storage media

Although PCMs are used as thermal energy storage media (TES-media) in several fields of engineering, such as those shown in Fig. 4; however, in this review, in addition to the storage area mentioned above in LHS units, special attention is also given, only for applications of PCMs as TES

storage media in the following areas: buildings, HVAC systems, cooling of electronic components, textiles, biomedical. Table 3 presents some recent studies regarding different applications of PCMs.

Remark 1 In the literature, several designations have been adopted for the PCM domain in an LHS storage unit, for example: PCM tank, PCM enclosure, PCM cavity, PCM container, PCM reservoir, PCM capsule etc. In addition, in the field of energy storage, the storage system is called an LHS unit (or also LHS module), in electronics the energy storage can be done in the cooling device, in the building sector, it represents, for example, the brick enclosure which is adapted to include a PCM inside, and so on.

Construction materials in buildings Energy consumption in buildings depends on commercial and residential activities. PCMs can significantly improve their energy performance. Indeed, when the building is heated, the PCMs (incorporated in its construction materials) melt and the energy is stored; while when it cools, the PCMs solidify and the energy is released. Generally, PCMs are integrated into buildings' construction materials by mixing them during manufacture or by encapsulation. In practice, there are several options for integrating phase change materials into building materials, for example: (1) inserting two PCMs into a brick cavity (enclosure); the first is encapsulated in the middle of the brick (Fig. 5a) and the second is encapsulated near the exterior brick wall (Fig. 5b) [29], (2) by embedding the PCM inside a brick enclosure and covering the inner wall of the brick (Fig. 3c) [30], and (3) by filling a glazed unit of the building with the PCM (Fig. 5d) [31]. Other examples of PCM-based construction materials in buildings adopted in the literature are given in Table 3.

HVAC systems In practice, there are several options for integrating phase change materials into HVAC systems, for example: (a) applying phase change material suspensions (PCMSs), consisting of microencapsulated phase change material suspension and clathrate hydrate suspension, in the secondary loop refrigeration and air conditioning system [32], (b) incorporating a PCM-based heat exchanger into a water-based HVAC system that uses *active* beams as terminal units [33], and (c) using two PCM storage tanks in the HVAC system: the first is used during the night to cool (therefore solidify) the PCM and the heat released by the solidification is then used to heat the building; while the second is used during the day to heat the PCM, the heat is therefore absorbed by melting which allows the building to be cooled [34]. A schematic representation of an LHS unit used in HVAC is given in Fig. 6. Other examples of PCM-based HVAC systems adopted in the literature are given in Table 3.

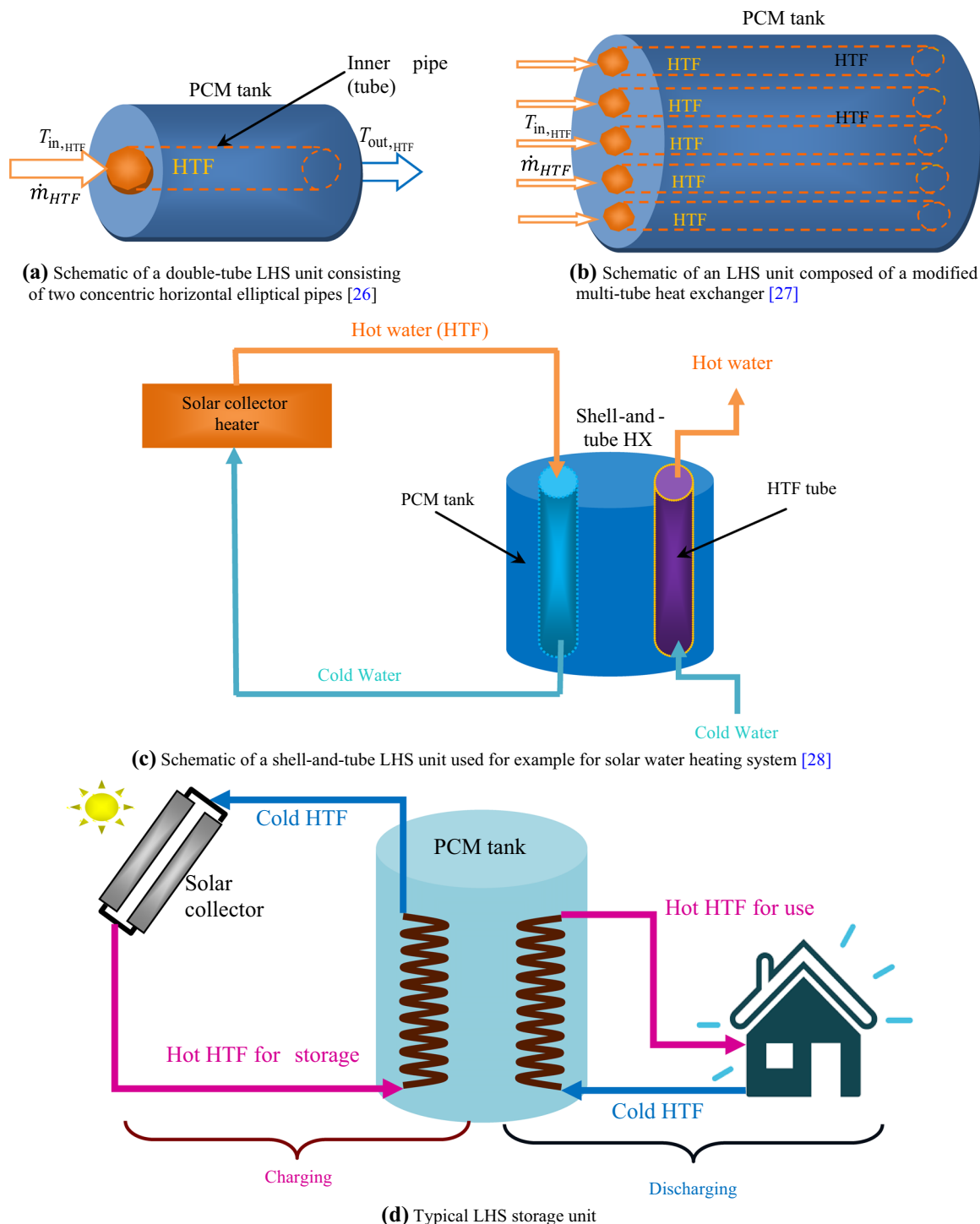


Fig. 3 Schematics illustrating some geometric shapes of PCM-based LHS units used in various TES applications. Other geometric shapes of LHS units are presented in Table 3

Electronic cooling Thermal control of electronic equipment plays a crucial role in its performance. Indeed, during operation, the electronic equipment heats up more and its conventional cooling has become ineffective; the discovery of new thermal control techniques is therefore of increased impor-

tance [35]. In this context, the use of PCM for electronic cooling is considered as the next generation of electronic cooling [36]. Several studies have been carried out in this area for example: (a) the use of a Ne-PCM (PCM enhanced by nanoparticles) dispersed around the cross section of an

Table 3 Some recent studies on PCM uses (applications)

References	Use/type of study	Description and parametric details	Main results
[56]	Energy storage tank, numerical and experimental	A TES tank using paraffin wax as PCM was studied. The objective was to predict the improvement in the overall performance (overall efficiency) of the installation.	It was found that the overall efficiency of the experimental installation improved and reached 62%.
[57]	Energy storage tank, experimental	Investigations were carried out in a TES system for different paraffin masses (mainly $m = 2, 4$, and 6 kg) and HTF flow rates.	It was found that the paraffin mass, $m = 2$ kg, showed better performance compared to other paraffin masses.
[58]	Energy storage tank, experimental	The objective was to predict the performance of the system. Multi-walled carbon nanotubes (CNTs) were dispersed in D-mannitol (DM) to prepare thermally conductive phase change nanocomposite materials (DM-CNTs), enhanced by mass fractions of 0.1 and 0.5 mass% of CNT, respectively. The objective was to predict the improvement in thermal conductivity, k .	It was found that for DM-CNT composites at CNT mass percentages of 0.1 and 0.5 mass%, the maximum improvement in thermal conductivity, k , was 7% and 32%, respectively. In addition, the total duration of a complete cycle (storage time) is reduced by 25.7% for the DM-CNT of 0.5 mass% composites.
[59]	Energy storage tank, numerical	The impact of design parameters, including HTF inlet temperature, HTF flow rate, PCM capsule size, and PCM initial temperature, on the thermal performance of a compact bed heat storage device, was studied and analyzed using a mathematical model. The objective was to analyze the thermal performance of packed bed phase change thermal storage.	It was found that increasing the HTF inlet temperature from 333 to 363 K resulted in a three-fold increase in energy destruction, a 67% decrease in effective heat storage duration, an increase of 38% in effective heat storage, and an 11% decrease in energy efficiency. In addition, the reduction in capsule diameter has a positive impact on all evaluation indices. The order of importance of the parameters (in their range of variation) is the inlet temperature of the HTF, the flow rate of the HTF, the size of the PCM capsule, and the initial temperature of the PCM.
[60]	Energy storage tank, experimental	A heat storage system filled with PCM enclosed in "sheets" using a liquid heat transfer fluid (HTF). The objective was to study the impact of the thickness of the PCM, the width of the space between the plates (sheets), and the mass flow rate on the performance of the heat storage system.	It was found that increasing the parameters, namely PCM distance (space between the PCM plates), HTF flow rate and slab thickness, reduced the energy storage efficiency.
[61]	Energy storage tank, numerical	A heat storage system for combined heat and power generation using microencapsulated phase change material slurry (MPCMS) as working fluid (HTF) was studied theoretically. The objective was to study the possibility of using microencapsulated phase change material slurry for energy storage.	It was found that the internal melting rate of microcapsule particles is gradually low; 90% of the latent heat can be absorbed (stored) in 63% of the total melting time. Furthermore, a 300-MW cogeneration case study showed that peak shaving capacity is increased by 81.4% through the use of an MPCMS thermal energy storage system.
[62]	Energy storage tank, experimental	Shape-stable phase change composite fibers were fabricated by the co-electrospinning method by combining binary eutectic fatty acids, capric acid-lauric acid (CA-LA), and capric acid-palmitic acid (CA-PA). The study aimed to encapsulate fatty acid eutectics in polyethylene terephthalate to obtain composite fibers with desirable thermophysical properties.	It was found that co-electrospun composite fibers offer lower melting temperatures and a wide range of phase transition temperatures (-9 to 34 °C) with high latent heat of fusion values.

Table 3 (continued)

References	Use/type of study	Description and parametric details	Main results
[63]	Energy storage tank, experimental	Using dodecanoic acid as an energy storage material and graphene as a supporting matrix, a shape-stable phase change composite material was prepared by in situ filling of carbon acetate (Na_2CO_3) cores from solid sodium acetate through an impregnation process	The enhanced energy storage capacity was found to be up to 101.4% higher than the expected energy storage capacity with the reduced supercooling phenomenon. And about 78% improved photothermal conversion and up to 159.1% thermal conductivity compared to pure PCM
[64]	Energy storage tank, numerical	An LHS unit using multistage PCM with different thermo-physical properties encapsulated in spherical capsules was studied The objective was to study the increase in heat transfer rate between HTF and PCM	It was found that increasing the number of PCM stages improves the performance of the energy storage system. However, this improvement is not significant when using more than three stages
[65]	Energy storage tank, numerical	A mathematical model based on the enthalpy formulation was developed to analyze the impact of PCM thickness on the temperature distribution and liquid fraction during the PCM melting process	It was found that the variation of melting time is linear with the amount of PCM Furthermore, for better PCM performance, a lower PCM thickness should be used
[66]	Energy storage tank, numerical	A novel double-tube latent heat storage unit was proposed The objective was to combine techniques to reduce the melting time in the proposed storage system	The proposed unit was found to reduce the melting time by 76%, 66%, and 53%, respectively, compared to vertical, horizontal, and helical-coiled storage systems
[67]	Buildings, numerical, and experimental	The use of paraffin for free cooling by storing the cold overnight for use for daytime cooling was investigated The study aimed to analyze the impact of paraffin (PCM) on building cooling	It was found that the use of 3.6 kg of paraffin makes it possible to cool the air below a temperature of 24 °C for 2.5 h and for an air velocity of 1 m/s
[68]	Buildings, numerical	The use of two PCM composites: “mixed-type PCM-gypsum” and “shape-stabilized PCM” plates in a passive solar building in Beijing was investigated The objective was to analyze the performance of the two PCMs	The optimal melting temperature was found around 21 °C. Furthermore, it was found that “mixed-type PCM-gypsum” and “shape-stabilized PCM” plates effectively reduced the internal temperature difference by 46% and 56%, respectively In addition, “Shape-Stabilized PCM” plates perform faster than “mixed-type PCM-gypsum” plates and are more thermally efficient in terms of latent heat utilization (storage)
[69]	Buildings, numerical	A new storage system based on the possibility of changing the position of the PCM layer relative to the insulation during the day so that it faces outwards at night and inwards during peak consumption hours was proposed The objective was to examine the impact of PCM position on the performance of the proposed system	It was found that the proposed system provides 379% lower cooling load (charge) than a system without PCM Furthermore, it was concluded that the innovative technology is designed not only for use in vertical walls but also for roofs and ceilings
[70]	Buildings, numerical	The incorporation of phase change materials (PCMs) into a typical office building in different climatic zones was investigated The objective was to predict the impact of PCM on energy consumption	It was found that the application of PCMs proved to provide greater energy savings for office buildings, regardless of climatic zones

Table 3 (continued)

References	Use/type of study	Description and parametric details	Main results
[71]	Buildings, experimental	Two identically sized experimental rooms were constructed and tested, one with PCM and the other without PCM	The thermal performance of lightweight buildings is greatly affected by seasons, and the contribution efficiency of PCM varies with seasons Compared with the reference wall, the attenuation rate of internal surface temperature can be reduced by 18.08% to 42.90% depending on the seasons, and the delay time can be improved by 2.67 to 4 h By successfully preventing indoor temperature fluctuations and increases, PCM can lower the maximum indoor temperature by 4.9 to 12.0 °C, increase the minimum temperature by 1.1 to 2.8 °C, and add 2 to 5 h of thermal comfort Lightweight buildings incorporating PCM can save 15.9% for winter heating and 18.69% and 49.63% for peak cooling in summer and transition, respectively Increasing airflow has been shown to result in high heat transfer rates and therefore high PCM thermal storage unit outlet temperatures In addition, the inlet temperature of the PCM thermal storage unit has been shown to have a significant impact on the discharge process Since the heat absorbed from the PCM into the unit during discharge; it was therefore suggested that this mechanism could be used to cool the area and precondition the air before it enters the control unit The HMPC controller can determine the optimal operating mode to maintain indoor comfort conditions within a defined thermal comfort range, which may change, and maximize overall energy efficiency using all resources and energy available on site In all studied cases, the coefficient of performance of the air conditioning system with PCM was higher than its coefficient of performance without PCM at a given time The shorter fusion times were caused by higher intake air velocity and Stefan number. In particular, with a fixed Stefan value and an outer diameter/inner diameter ratio value of 2, the time required for complete fusion decreased by 19.16% with an increase in velocity from 1.6 to 2.14 m/s The time decreased by 31.59% when the velocity is increased to 2.67 m/s. For low velocity cases, better coefficient of performance values were found Nevertheless, the total fusion time increased from 3.35 to 7.11 h due to the increase in PCM volume
[72]	HVAC, numerical and experimental	An HVAC system consisting of an onboard photovoltaic (PVT) collector and a phase change material (PCM) collector integrated with a reversible heat pump in a ducted system was studied The objective was to examine the performance of the HVAC system	
[73]	HVAC, experimental	A hybrid model predictive control (HMPC) system implemented to control a new solar-assisted HVAC system was studied The objective was to evaluate the effect of using PCM on the performance of the new HVAC system By analyzing the effects of crucial parameters including inlet velocity and Stefan number on melting rates, outlet temperature, and coefficient of performance, the impact of using a phase change material (PCM)-based latent heat storage (LHS) combined with an air conditioning system was evaluated	
[74]	HVAC, experimental		

Table 3 (continued)

References	Use/type of study	Description and parametric details	Main results
[75]	Electronic cooling, experimental	The incorporation of a phase change material (PCM) into the heat sink (HS) was investigated The objective was to examine the thermal performance of PCM for its use as a thermal interface material in HSs	It has been found that when the temperatures to which PCM is subjected exceed 150 °C, it begins to decompose and lose mass until it loses 13% of its organic components
[76]	Electronic cooling, numerical and experimental	The integration of a phase change material into a heat sink to study its impact on its cooling performance was examined The objective was to examine the thermal performance of PCM for its use in HSs	It has been found that using a heat sink based on phase change material (PCM) improves cooling performance compared to a heat sink without PCM
[77]	Electronic cooling, numerical and experimental	Covering the adiabatic cross section of a heat pipe (HP) using a storage tank using tricosane as a phase change material (PCM) was investigated The objective was to evaluate the cooling performance of heat pipes (HPs) using PCM	It was found that using PCM can reduce fan power consumption by 46% compared to traditional heat pipes
[78]	Electronic cooling, experimental	The use of paraffin wax or 1-hexadecanal as PCMs to cool a heat sink (HS) under constant and cyclic charging by natural and forced convection was investigated The objective was to predict and compare the cooling system performance using these PCMs	It has been found that the charging cycle provides better heat sink performance. For cycle charging, the use of 1-hexadecanal reduced the cooling time compared to paraffin When the heat sink is cooled by forced convection, the operation time is reduced by 0.41 and 0.43 times compared to natural convection for paraffin and 1-hexadecanal, respectively
[79]	Electronic cooling, numerical	PCM combined with triply periodic minimal surface (TPMS) designs were used. In comparison with pure PCM heat sinks, three PCM-TPMS heat sinks (PCM-Gyroid, PCM-IWP, and PCM-Primitive) are investigated The aim was to study the thermal management of electronic components	Compared with pure PCM heat sinks, PCM-TPMS heat sinks (PCM-Gyroid, PCM-IWP, and PCM-Primitive) improve heat transfer and significantly reduce core (base) temperatures, averaging 61.9 °C and 34.9 °C in passive and active cooling, respectively The temperature reduction is somewhat improved by improving the relative density of the TPMS by 10% to 30%; the PCM-Gyroid layout is the most efficient In addition, especially thermal cycling, PCM-TPMS architectures increase the reliability and longevity of electronic components
[80]	Textiles, experimental	The use of the composite formed from melamine-formaldehyde microencapsulated with eicosane (PCM) in a textile was studied The objective was to examine the effect of this composite PCM on the thermoregulation of the textile	It was found that the heat storage capacity increases with the concentration of the microcapsules. Indeed, it has been shown that thermoregulatory tissues have capacities of 0.91 to 4.44 J/g, depending on the concentration of the microcapsules Treated textiles have also been shown to retain 40% of their heat storage capacity after five washes

Table 3 (continued)

References	Use/type of study	Description and parametric details	Main results
[81]	Textiles, experimental	Two PCMs, capric acid and stearic acid, were directly added to cotton in the presence of magnesium chloride and a hydroxymethyl-poly-siloxane binder (without microencapsulation) to improve the performance of treated cotton	It was found that direct application of 5 vol% capric acid and 5 vol% stearic acid to cotton results in a phase change at 17 °C, 29 °C and 49 °C, 60 °C for cooling and heating cycles, respectively Furthermore, direct application of 5 vol% capric acid and 5 vol% stearic acid to cotton indicated that a double phase change occurred separately at 17 °C and 30 °C for cooling and heating “capric acid” and “stearic acid,” respectively; and at 51 °C and 60 °C for the cooling and heating cycle of “stearic acid,” respectively It was found that the fabric treated with 22.9 vol% additional microcapsules was able to absorb 4.44 J/g of heat In addition, air and water vapor permeability was shown to decrease, while moisture absorption increased with increase in additives In addition to having low thermal conductivity (10.4 mW/m K) and high fusion enthalpy (74.27 J/g), 2D phase change fiber textiles have a stable structure. They are suitable for body temperature control because they effectively delay temperature changes After fatigue tests, 3D textiles made from phase change composite fibers exhibit increased tensile strength and can support up to 4,000 times their weight. With a thermal conductivity of 14.1 mW/m K, they provide superior thermal insulation compared to commercial cotton and can easily recover their shape after compression It was found that PCM has a nonsignificant effect on the phase transition temperature of the poly(methyl-methacrylate) matrix and that the thermal regulation capacity propagates around body temperature, by absorbing or releasing a thermal energy of up to 30 J/g It was found that the inclusion of micro/nano PCM can decrease the temperature of healthy tissues and has no obvious impact on the temperature of cancerous tissues under the same heating conditions, which can help protect healthy tissues during hyperthermia
[82]	Textiles, experimental	The polyester knits were treated with “PCM microcapsules” using the “buffer drying process” with “polyurethane binder” to evaluate the influence of “PCM microcapsules” on the heat absorption capacity (heat storage capacity) and the water vapor permeability of these fabrics Direct electrospinning of a polystyrene/polyethylene glycol blend was used to create 2D textiles containing phase change fibers To improve their mechanical properties, multi-needle electrospinning was used to fabricate 3D textiles with composite phase change fibers	
[83]	Textiles, experimental		
[84]	Biomedical, numerical	A microencapsulated paraffin-based phase change material (PCM) was incorporated into a poly(methyl-methacrylate) (PMMA) matrix to predict the ability to store and release heat around body temperature	
[85]	Biomedical, numerical	A novel thermal protection method using microencapsulated phase change microparticles/nanoparticles during hyperthermia was proposed to predict the impact of PCM on thermal protection during hyperthermia	

Fig. 4 Overview of the main application areas of PCMs: It should be noted that apart from the areas presented, PCMs find their application in refrigeration, batteries, solar concentrators, solar water heating systems, clean energies, space exploration, thermal conditioning of products agricultural, etc.

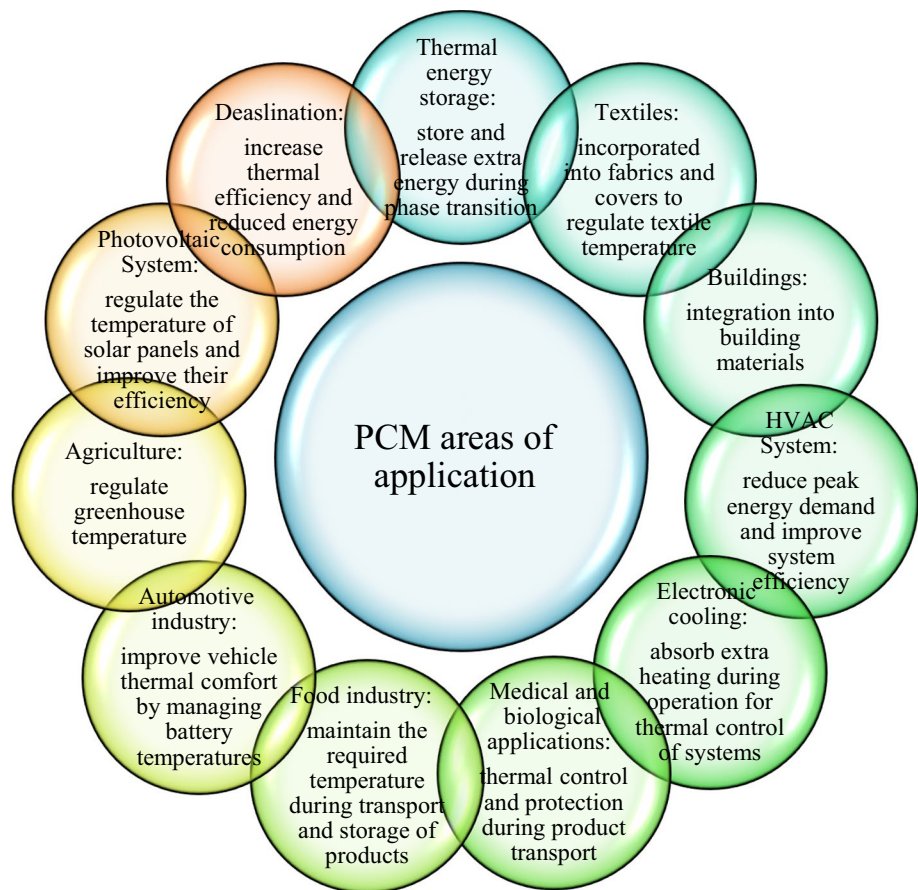
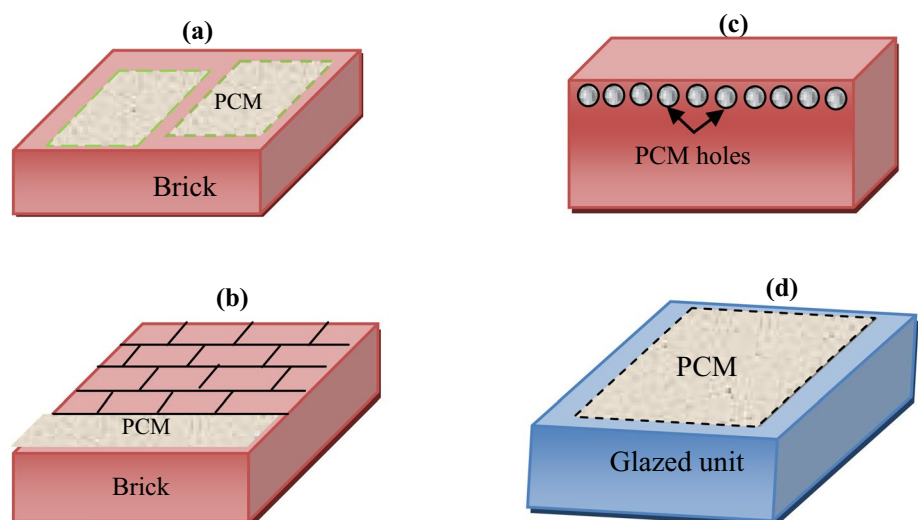


Fig. 5 Examples of how to incorporate PCMs into construction materials [29–31]



adiabatic heat pipe (HP) (see Fig. 7a) [37], (b) the use of composite foam-PCM as a heat sink (HS) (see Fig. 7b) [38], and (c) the incorporation of a PCM into three-dimensional oscillating heat pipes (3D-OHP) (Fig. 7c) [39]. Other examples of PCM-based heat sinks adopted in the literature are given in Table 3.

Textiles The technology to use phase change materials in textiles was developed by NASA in the early 1980s. It was used to protect astronauts from the extreme temperature variations they encountered in space [40]. PCMs are used in textiles to improve their thermal performance by varying the aggregation state within a defined temperature range

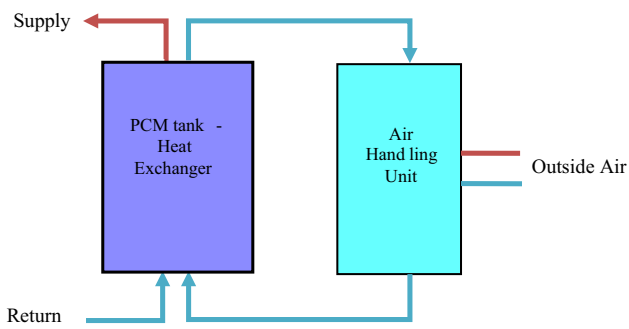


Fig. 6 Schematic representation of an LHS unit used in HVAC [32–34]

(see Fig. 8) [41]. Textiles were one of the first applications of PCMs. A review of PCMs for textile applications was carried out by Arjun et al. [42]. Furthermore, various studies in the literature have been devoted to this area of use of PCMs, for example: Covering and padding of staple and crimped polyester fibers to improve their performance [43]. Treatment and painting of cotton fabrics with PCM microcapsules for commercial use as active sportswear [44]. Microencapsulation of knitting with homogeneous PCM microcapsules and their mixtures [45]. Other examples of PCM-based textile technologies adopted in the literature are given in Table 3.

Biomedical PCMs are currently used for biomedical applications ranging from thermal control of biological tissues during transportation/storage to thermal control/protection for in situ operations, as shown in Fig. 9. Various studies in the literature have been devoted to the area of use of PCMs in the biomedical field. For example: (a) thermal control during transport of organic products, Mondieig et al. [46], and (b) protection of healthy tissue surrounding cancerous tissue during cryosurgery, Lv et al. [47]. Other examples of PCM-based biomedical fields adopted in the literature are shown in Table 3.

Food and agriculture industries Phase change materials (PCMs) have several promising applications in food and

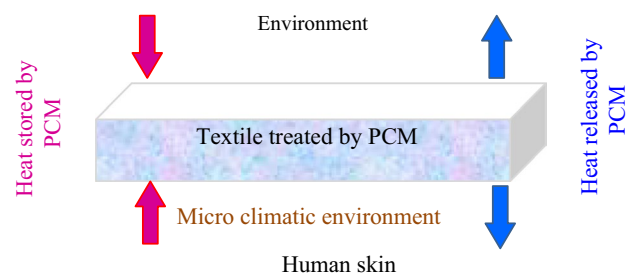


Fig. 8 The use of PCM in textiles [41]

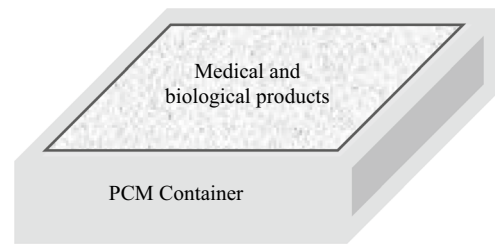


Fig. 9 The use of a PCM in biomedical

agriculture. They can be used in (a) insulated boxes or containers to improve temperature control during food storage and transportation (see Fig. 10a) [48], (b) incorporated into materials used for greenhouse roofs and walls (see Fig. 10b) [49] for cultivation to control temperature, improve plant development, and increase overall energy efficiency, (c) incorporated into the root zone of the greenhouse to help control soil temperature (see Fig. 10c) [50], and (d) incorporated into food drying equipment (see Fig. 10d) [51] to improve product quality, increase energy efficiency, and make the drying process more sustainable.

Automotive The performance and thermal comfort of vehicles can be improved while increasing energy savings by integrating PCMs for thermal control in the automotive industry. Indeed, PCMs can be integrated into a variety of automotive components, such as (a) around battery cells or

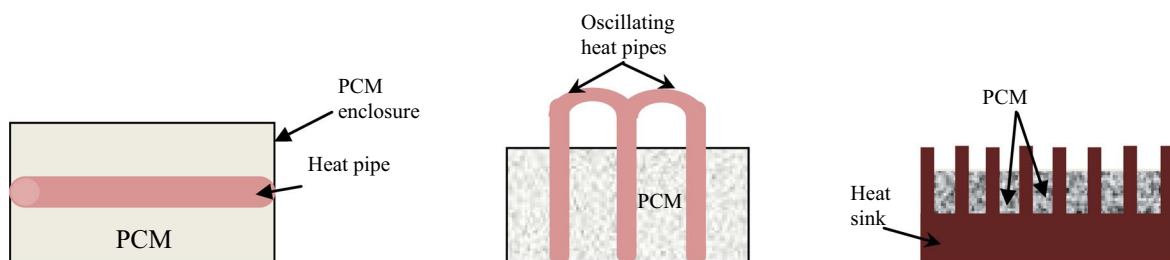


Fig. 7 Examples of PCM-based HPs and HSs used in electronic cooling, adapted, respectively, from [37–39]

inside battery boxes (see Fig. 11a) [52] and (b) in the roof of a vehicle (see Fig. 11b) [53].

Desalination The low efficiency of solar desalination systems can be addressed by using phase change materials (PCMs) in desalination, which can lead to increased thermal efficiency, reduced energy consumption, and improved

overall performance. PCMs can be strategically included in desalination systems in several ways, (a) separately encapsulated in the basin (see Fig. 12a) [54] and (b) under the evaporator basin (see Fig. 12b) [55].

Fig. 10 The use of PCMs in the food and agriculture industries [48–51]

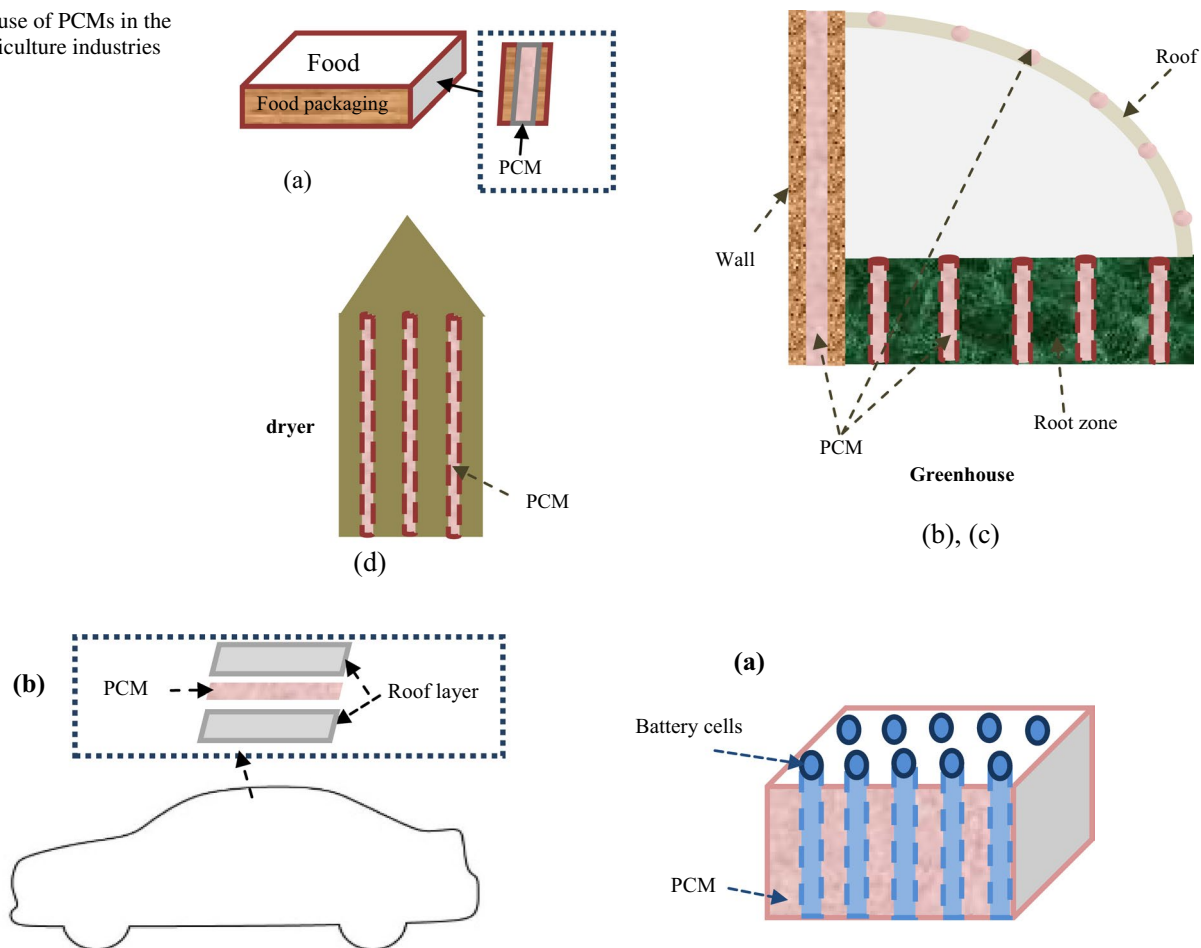


Fig. 11 The use of a PCM in automotive [52, 53]

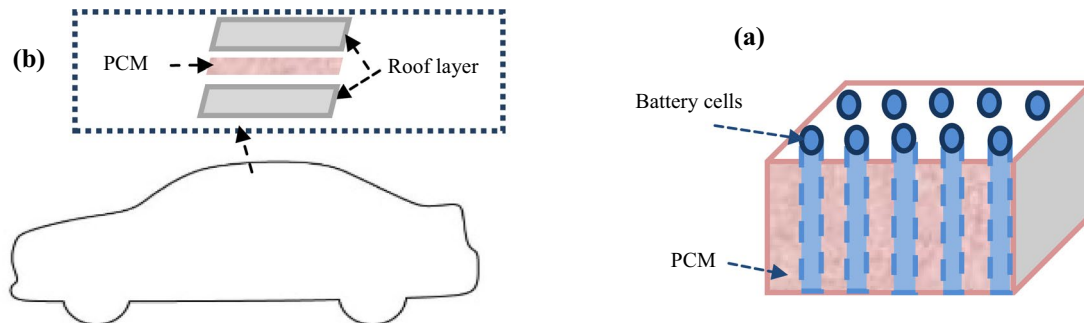
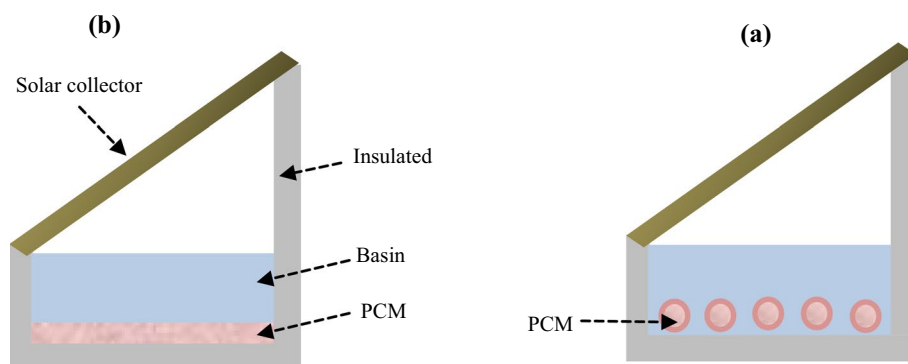


Fig. 12 The use of PCMs in desalination [54, 55]



Discussions on the classification and use of PCMs

As mentioned earlier, PCMs are substances that can absorb or release thermal energy during their phase transition from solid to liquid or vice versa, respectively. They are classified according to their physical state and nature into three categories: organic, inorganic, and eutectic. These PCMs are used in various applications including latent heat storage units, buildings, HVAC systems, electronic cooling, the textile industry, and the biomedical field, etc.

Use of PCMs in thermal energy storage systems (LHS units)

The latent heat storage system (LHS unit) is generally based on the use of a tubular heat exchanger which ensures the heat exchange between the HTF fluid and the PCM (see Fig. 3). This exchange depends on different parameters, such as the temperature of the HTF at the inlet of the inner tube (i.e., the temperature at the start (initial) of the charging or discharging process), $T_{in,HTF}$, and the flow rate of the HTF, ($\dot{m}_{in,HTF} = \dot{m}_{out,HTF} = \dot{m}_{HTF}$), as well as the thermophysical properties of the storage medium (the PCM). According to the literature reviewed here, the use of PCMs with appropriate properties increases the volumetric heat storage capacity of the storage unit. In addition, increasing the temperature of the HTF at the inlet of the inner tube and the flow rate of the HTF can also improve the energy stored in the PCM, hence the volumetric heat storage capacity of the storage unit.

Use of PCMs in buildings and HVAC systems

Heating and cooling of buildings using PCMs constitute a very important application sector for PCMs. The application relies on mixing PCMs with building materials encapsulating them inside bricks or embedding them between layers of bricks in walls or ceilings. The principle of this application is based on the absorption of heat by the PCM during its melting for the cooling needs of the building. Then, this absorbed (stored) heat is released by the PCM as the PCM solidifies for the heating needs of the building. This technique therefore makes it possible to regulate the temperature inside the building, reduce heat loss and thus increase energy savings. When talking about buildings, we should also mention HVAC systems, the efficiency of which can be improved through the use of PCM. This is because, during off-peak periods, the HVAC system's PCM melts and absorbs energy, which is then released (upon solidification) when needed. This can reduce peak energy demand, matching energy demand and supply, and improving the overall efficiency of the HVAC system.

Use of PCMs in electronic cooling (PCM-based heat sink devices)

For electronic cooling, PCM is used to absorb heat generated by electronic components during intense device activity to ensure a more stable temperature through advanced thermal management to prevent device overheating. The technique relies on the thermal absorption of the equipment. This is because as the temperature increases, the PCM melts and absorbs a significant amount of heat, which is stored as latent heat and keeps the equipment temperature constant. The use of this technique involves (i) embedding PCM capsules or PCM materials directly into the heat sink structure, (ii) applying them as a coating around heat-sensitive electronic components, or (iii) integrating PCMs into the materials that make up the housing of electronic equipment.

Use of PCMs in the textiles industry

The application of PCM in textiles is established either in textile fibers or through a flexible method involving microencapsulation in different parts to improve the thermal comfort of clothing and adapt to different environments. They thus ensure a constant body temperature by absorbing body heat when the temperature rises and releasing it when it falls. In addition, the use of PCM in textiles can reduce humidity. The technique relies on the absorption of heat by the PCM when the environment is hot and the temperature is above the melting point of the PCM; the PCM absorbs heat and stores additional (extra) heat as latent heat without any change in body temperature. As the ambient temperature decreases, stored latent heat is released into the environment, keeping the body temperature stable and comfortable. This technique is used by (1) applying PCM to the surface of textiles by coating or impregnation, (2) incorporating PCM into the fibers of the textile, (3) adding PCM-containing coatings to the fabric, or (4) directly integrating PCM into textile fibers during the fibers manufacturing process.

Use of PCMs in the biomedical field

The biomedical field, particularly in the transport of drugs and the storage of biological products while maintaining stable and comfortable temperatures, represents a promising alternative to PCM applications. Examples of applications of PCM in this area include (a) integration of PCM into drug delivery systems to control drug release rate based on temperature changes, (b) inclusion of PCM in the design of composite materials for medical implants, (e.g., dental or orthopedic implants), to help regulate the temperature around the implants, and (c) the integration of PCM into storage and transport systems for biological products, such as transport containers or freezers, to maintain a stable

temperature and avoid thermal fluctuations that could damage (harm) the product.

Use of PCMs in food and agriculture industries

In the agriculture and food industries, phase change materials (PCMs) are used to improve productivity and temperature control. By regulating the temperature in storage facilities and shipping containers, they help extend the shelf life of food and prevent spoilage. In logistics, they regulate the temperature throughout transportation. To promote the best possible plant growth in greenhouses, PCMs absorb excess heat during the day and release it at night. They can also increase the efficiency of refrigeration systems in the food industry and lower soil temperatures, improving product quality, reducing waste, and increasing energy efficiency. In addition, PCMs can be used to maintain constant temperatures by storing excess heat generated during the drying process and gradually releasing it.

Use of PCMs in automotive

Thermal performance and comfort in vehicles can be improved while increasing energy savings by integrating PCMs for thermal control in the automotive industry. To improve their thermal performance and stop leaks, PCMs are often enclosed in containers called microcapsules. PCMs can be integrated into several automotive parts, such as battery cases or surrounding cells to maintain ideal temperatures, seat or upholstery covers to help control cabin temperatures, and spaces between layers of vehicle body panels to control heat absorption.

Use of PCMs in desalination

Phase change material (PCM) modules integrated into modular desalination systems are an effective way to increase water production. To maintain ideal temperatures, each module is composed of layers of PCM that collect heat during the day, promoting the evaporation of seawater, and release this stored energy at night. In addition to increasing water production and thermal stability, this design allows for easy expansion and customization based on demand. These systems use PCM technology to save energy and operating costs, making them a sustainable choice to solve the water crisis.

Challenges and limitations of using PCMs

Despite all these advantages of PCMs, it should be emphasized that the use of PCMs also requires consideration of the challenges and limitations associated with their use, including cost, appropriate choice of PCM for a specific

application, management of durability of PCM and optimal design and system integration using PCM (see also the disadvantages of PCMs provided in Table 2 above).

Phase transition processes

Definition and principle of a phase transition

Phase transition processes of a PCM can be performed in the following forms: solid–solid, solid–liquid, gas–liquid, and gas–solid. The transformation of a phase change material from one state to another is a nonlinear transient phenomenon that occurs with the presence of a moving solid–liquid interface (also called a mushy zone, for example [86]). This problem is generally referred to as the *moving boundary* problem [87]. In the present article, attention is paid only to solid–liquid and liquid–solid transitions which are the main phase transformations that can occur in the field of energy storage. Figure 13 gives a schematic representation of these two phase transitions (transformations).

Recall that phase change materials (also called latent heat storage materials) are specifically used in latent thermal energy storage systems, as discussed above. They are capable of absorbing (i.e., storing) (during melting) and releasing (during solidification) large amounts of latent heat over a near-isothermal temperature range (interval) during phase transition [88]. Many studies have been published regarding the melting and solidification of PCMs and are listed and analyzed in Table 4.

Discussion of phase transition processes

Melting process

An analysis of the studies presented in Table 4 indicates that during the initial stages of the melting process, conduction dominates the heat transfer and after that, convective

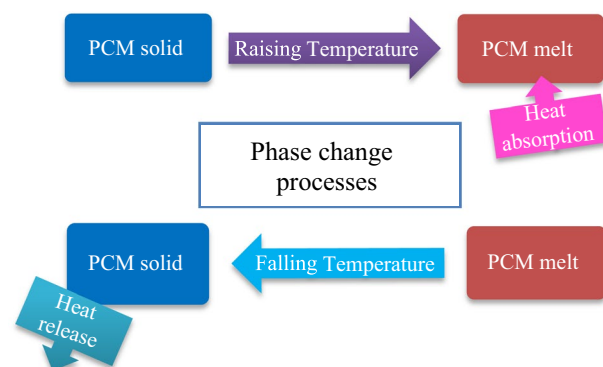


Fig. 13 Illustrating scheme of the phase transitions

Table 4 Summary of some recent studies regarding PCM's melting and solidification process

References	Process/type of analysis	Description and parametric details	Main results
[112]	Melting, numerical	The melting of a PCM within a rectangular enclosure subjected to different heat transfers by convection ($h = 5, 10$ and $15 \text{ W/m}^2\text{K}$) and at different ambient temperatures ($T_\infty = 297$ and 307 K) was studied The objective was to predict the impact of convective heat flux on PCM melting in this latent heat storage system	It was found that the melting rate increases with increase in heat transfer coefficient h when T_∞ is higher than the melting temperature, and the melting rate decreases when T_∞ is equivalent to the melting temperature
[113]	Melting, numerical	The melting of RT82 (PCM) in an LHS unit, consisting of a quadruple-tube heat exchanger (QTHX), was studied sequentially using two different methods (processes). The first process is when the PCM is included in the first tube and the third tube. The second process is when the PCM is included in the second tube and the fourth tube The objective was to evaluate the effectiveness (efficiency) of these methods. The study also compared the results of this process with the cases of triplex tube heat exchanger (TTHX) and double-tube heat exchanger (DTHX) LHS units	It was found that the melting rate of the second process was achieved at 63.2% of that of the first process In addition, the comparison showed that PCM melted earlier in the QTHX unit than in the TTHX and DTHX units
[114]	Melting, experimental	The melting of a bio-based phase change material in a rectangular latent energy storage unit was studied using various porosities of full-size wire mesh and partial wire mesh at different locations The objective of the study was to improve the charging process	It was shown that the uniformity of temperature distribution was reduced with the addition of wire mesh In addition, full-size wire mesh with porosities of 0.88 and 0.82 was found to reduce charging time by 41% and 52%, respectively, while a partial wire mesh with 0.91 porosity positioned at the bottom and top reduces the charging time by 34% and 7%, respectively
[115]	Melting, experimental	The melting process of paraffin-based PCM into rectangular high-density polyethylene panels was studied. Different temperature and air velocity values as well as different panel inclination angles were tested to examine their impacts on melting performance (especially melting time)	Increasing air temperature, air velocity, and panel inclination angle have been shown to improve the melting rate Moreover, the melting time reduction was shown to be between 43 and 47%, 36.6% and 49.2%, and 18.8% and 50.8% when air temperature, the air velocity, and inclination angle were increased from 35 to 55 °C, 1 to 4 m/s, and 30° to 90°, respectively. However, when the air velocity increased from 4 to 5 m/s, the reduction in melting time was reduced from 49.2% to 46.7%, respectively
[116]	Melting, numerical and experimental	The thermal coupling between the heat transfer fluid (HTF) and the phase change material capsule inside a thermal energy storage unit was modeled numerically and studied experimentally The objective was to compare numerical and experimental experiments	It was found that the results of the numerical model are very consistent with the experimental results

Table 4 (continued)

References	Process/type of analysis	Description and parametric details	Main results
[117]	Melting, numerical	The melting process of n-octadecane (PCM) in a storage unit consisting of circular sections and having two shapes called “half geometry” and “up geometry” and subjected to the action of thermocapillary was studied The objective was to evaluate the effect of thermocapillarity and Stefan number (for two Stefan number values 0.22 and 0.67) on the melting process	Thermocapillarity was shown to decrease the total melting time by a factor of 3.6 (Ste=0.22) and 5.23 (Ste=0.67) for the half geometry and by a factor of 3.2 (Ste=0.22) and 5.18 (Ste=0.67) for the superior geometry of the storage unit compared to melting by conduction alone (absence of thermocapillarity)
[118]	Melting, experimental	The melting of RT44Hs PCM in a test cell (storage unit) of rectangular shape made of a cross section of polycarbonate sheet, copper plates, and mica heaters was studied. The cell was subjected to different heat flux values (675, 960, and 1295 W/m ²), imposed on its left and right sides (walls) The objective was to analyze the impact of heat flux boundary conditions on the melting rate	Increasing the heat flux from 675 W/m ² to 960 W/m ² and 1295 W/m ² was shown to reduce the total duration of the melting process by 27.27% and 43.18%, respectively
[119]	Melting, numerical and experimental	The melting processes of pure paraffin PCM as well as aluminum-paraffin composite PCM in storage units were studied numerically and experimentally The objective was to evaluate the difference between pure PCM and composite PCM melting processes in terms of performance	The experimental results showed that the temperature difference (10.04 °C) is smaller in pure PCM, while the numerical results showed that it is smaller (5.18 °C) in the composite PCM Furthermore, it was found that the melting of pure PCM is dominated by natural convection; while the melting of composite PCM is dominated by conduction It was concluded that the numerical results were consistent with the experimental results
[120]	Melting, experimental	The melting processes of “pure paraffin PCM” and “saturated paraffin PCM in copper foam” in a storage unit (cavity) with different inclination angles (30°, 60°) The objective was to predict the effect of storage unit inclination on melting time	It is shown that by using an inclination angle of 30°, the total melting time of pure PCM is reduced by approximately 25.30% compared to the 60° case In addition, it was found that different inclination angles have little effect on the melting process of PCM composite metal foam
[121]	Melting, numerical	A new PCM container (storage unit) with an elongated and curved shape based on the guidelines was proposed to improve the convection and hence the melting process. This container design was compared to a standard rectangular container to evaluate the effectiveness of this new container	It was found that the increase in natural convection velocity caused by the new container design improves the PCM melting process Furthermore, it was found that the melting time of the new container was reduced by more than 20% compared to the rectangular container

Table 4 (continued)

References	Process/type of analysis	Description and parametric details	Main results
[122]	Melting, numerical and experimental	A new numerical model employing an implicit algorithm, based on an extended Darcy term, was developed to study the melting process of a filled PCM in a cubic enclosure-shaped storage unit The numerical model was compared to other existing numerical models (based on explicit algorithms) and validated by experiments in order to evaluate its ability to study the melting performance of PCM	It was found that the use of an implicit algorithm significantly improves the stability of the numerical model Furthermore, it was found that the sedimentation velocity of PCM is almost stable only during a certain time of the melting process, and the melting time decreases when the wall temperature is higher
[123]	Melting, numerical	The process of melting of a PCM included inside a capsule (storage unit) "called PCM capsule," which is exposed to natural convection and crossed by a heat transfer fluid (FHT) circulating outside of this capsule, has been studied The objective was to analyze the impact of the size of the PCM capsule and the characteristics of the HTF at the capsule inlet on the melting process	It has been shown that the melting rate increases with decrease in capsule size and that the effect of natural convection is significant for larger capsules with a 27.04% decrease in melting time for a capsule of radius 6 mm In addition, increasing the HTF inlet temperature and velocity has been shown to improve the melting rate Finally, it was found that the melting rate decreased by about 15.04% and 13.88% with natural convection for flow from the bottom and top, respectively
[124]	Melting, numerical and experimental	The melting of PCM paraffin wax in a cylindrical storage unit was studied The objective was to determine the effect of three different inclination angles 0°, 45°, and 90° on the melting process	It was found that the melting rate was higher at the 45° angle than at the 0° and 90° angles
[125]	Melting, numerical	Mathematical and numerical models were developed to study the melting process of a PCM in a storage unit (rectangular cavity) The objective was to examine the effect of temperature imposed on the bottom wall cavity on PCM melting	It has been shown that the melting process can be controlled by changing the temperature in the storage cell (cavity) Furthermore, it has been demonstrated that the proposed numerical method has good accuracy
[126]	Melting, numerical	The gallium (PCM) melting process in a rectangular storage unit (cavity) heated from its vertical wall was examined using numerical discretization schemes (models) The objective was to perform an in-depth comparison between three common parameter schemes and multiple grid sizes	It has been shown that the problem of the size of the grid and the discretization scheme is actually a problem of convergence of numerical solutions. Indeed, the structure of the flow becomes multi-cellular as expected when the solution converges
[127]	Solidification, numerical and experimental	A model using temperature-dependent properties was developed to study the solidification of water PCM	It was found that the temperature difference between the experimental and numerical results obtained by another model in the literature was approximately 15 °C; while after using the newly developed model, this difference was reduced to around 1–2 °C

Table 4 (continued)

References	Process/type of analysis	Description and parametric details	Main results
[128]	Solidification, experimental	A method which consists of adding thermal energy conducting medium at the nanoscale (nanoparticles) into polyethylene glycol (PEG) (PCM) to reduce its porosity has been proposed	It was found that the inclusion of a conductive medium can reduce the porosity of PCM during phase transition. In particular, the use of PEG- Al_2O_3 with mass percentages of 5, 10, 15, and 20 mass% (of Al_2O_3) increases the thermal conductivity of the composite to 0.31, 0.39, 0.45, and 0.51 W/m K compared to that of pure PCM (PEG) which does not exceed 0.22 W/m K Likewise, it was found that the use of PEG- Al_2O_3 , PEG- CuO , and PEG- CuO leads to (i) at a cooling rate of 5°C m^{-1} the solidification temperatures increase, respectively, by 0.04, 4.69, and 2.58°C compared to the case of pure PEG, and (ii) at the cooling rate of 20°C m^{-1} the solidification temperatures increase, respectively, by 1.27, 4.28, and 4.12°C , compared to the case of pure PEG On the other hand, for pure PEG, PEG- Al_2O_3 , PEG- CuO and PEG- CuO which have latent heats of 154.6, 141.2, 154, and 151.8 J/g, respectively, the “molar stored energies” by these substances are found to be 207.9, 174.6, 169.6, and 282.7 kJ/mol, respectively It has been reported that increasing the Si content increases the crystallization (solidification) temperature. In particular, when the Si content (concentration) is 0, 10, 15, 20, and 25 mass%, the increase is 140°C , 175°C , 200°C , 220°C , and 240° , respectively In both materials, crystallization was shown to begin with the nucleation of small grains. Indeed, with the same heating rate (7.5°K/s), AgInSbTe starts to nucleate at 250°C ; while $\text{Ge}_2\text{Sb}_2\text{Te}_5$ starts at 170°C In addition, $\text{Ge}_2\text{Sb}_2\text{Te}_5$ has a nucleation rate of $8 \times 10^{14} \text{ cm}^{-3} \text{ s}^{-1}$; while AgInSbTe has a nucleation rate of $5 \times 10^{14} \text{ cm}^{-3} \text{ s}^{-1}$ It was demonstrated that the solidification rate was reduced by increasing the thermal resistance of the solid layer. In addition, decreasing the diameter increases the total solidification time; while increasing the temperature difference decreases it It was indicated that the freezing process within the Hele-Shaw cell is identical to that of a free sessile drop
[129]	Solidification, experimental	The crystallization of the phase change material Si-Sb ₄ Te ₄ with different silicon Si contents, was studied experimentally	
[130]	Solidification, experimental	The crystallization of the two phase change materials $\text{Ge}_2\text{Sb}_2\text{Te}_5$ and AgInSbTe was examined using a transmission electron microscope	
[131]	Solidification, numerical	The solidification of spherical erythritol capsules (PCM) of 10, 20, 30, and 40 mm diameter confined in an envelope with variable wall temperatures mainly 10, 15, 20, 25, 30, and 40 K lower than the temperature of solidification of erythritol was studied	
[132]	Solidification, experimental	The solidification of supercooled water (PCM) in contact with a solid wall using the Hele-Shaw arrangement (cell) was analyzed	

Table 4 (continued)

References	Process/type of analysis	Description and parametric details	Main results
[133]	Melting and solidification, numerical and experimental	The melting and solidification processes in two experimental PCM storage containers with and without internal fins were studied To model these processes, two methods, the first is “the enthalpy method” and the second is “the effective heat capacity method,” were used	The effective heat capacity approach with a narrow temperature range is the most accurate compared to experimental results
[134]	Melting and solidification, numerical	The processes of melting of 95pure “lauric acid” and freezing of water driven by buoyancy forces were studied using a <i>fixed-grid solution method</i>	It has been demonstrated that for Rayleigh numbers (Ra) in laminar flow mode a phase change can occur between fixed temperature boundaries (walls) It was also found that the numerical results are in agreement with the experimental data. Thus, it was concluded that the considered fixed-grid solution can be used to capture the physics of buoyancy-induced melting and freezing
[135]	Melting and solidification, numerical	The freezing and melting processes of water and ice PCMs surrounding horizontal heating and cooling cylinders in a square cavity of length H were analyzed. Three cylinder positions at $h = H/3$, $H/2$ and $2H/3$ from the bottom of the cavity were considered	The significant effect of superheating and supercooling on freezing and melting, respectively, during the first phase of both processes was demonstrated. This effect therefore gradually diminishes as the process progresses In addition, increasing the distance h significantly reduces melting, while freezing was more intense at $h = H/2$ compared to other locations that delayed freezing
[136]	Melting and solidification, numerical	The assembly consisting of a micro-foam embedded in a “phase change material” was placed in a container having an upper wall in which square waveform inlet temperature profiles are imposed The “micro-foam and PCM” assembly was subjected to different melting and freezing strengths and times. The so-called “one- and two-temperature volume averaging” and “direct simulation” models were used in the analysis	It was found that the average temperature and volume fraction of the molten phase change material predicted from the “average volume simulation” model are in good agreement with those predicted from the “direct simulation” model Likewise, both models did not accurately estimate the temperature difference between the micro-foam and the phase change material and were able to predict the heat flux of the top wall. Also, it was indicated that the maximum temperature difference between micro-foam and PCM increases with increase in melting and freezing strength or time
[137]	Melting and solidification, numerical	A numerical model based on an adaptive finite element method was used to simulate the solidification and melting processes of n-octadecane (PCM) in a square cavity	It has been found that solidification is generally slower than melting. Furthermore, heat transfer by convection dominates the melting process; however, conduction dominates heat transfer during the solidification process
[138]	Melting and solidification, experimental	The melting and solidification of an organic PCM without and with the addition of different mass fractions of aluminum (Al) powder were studied The whole is included in a sphere, first in hot water for melting, then in cold water for solidification	It was found that the PCM located at the top of the sphere melts faster than the one located at the bottom. However, the PCM near the inner wall of the sphere freezes faster than at the center of the sphere. In addition, when the initial PCM temperature is higher, melting is faster On the other hand, it has been shown that the presence of aluminum powder can accelerate the heat transfer of PCM during the melting and solidification processes

Table 4 (continued)

References	Process/type of analysis	Description and parametric details	Main results
[139]	Melting and solidification, experimental	The melting and solidification processes of PCM in an LHS storage unit with different oil flow rates and with perforated separator plates A and B installed in the PCM region were analyzed	The phase change time has been shown to decrease with increase in oil flow rate. In addition, the amount of heat stored without and with the perforated separation plate A is constant, while it is increased when perforated partition plate B is installed
[140]	Melting and solidification, numerical	The application of organic phase change materials for thermal conditioning agricultural products and the examination of the ability of PCM to regulate temperature were investigated	The use of PCM has been shown to improve temperature control of agricultural products and reduce transportation costs
[141]	Melting and solidification, numerical	Melting and solidification in a latent heat thermal storage system using in two, three, and five PCMs were considered	It was found that the use of multiple PCMs improves significantly energy efficiency
[142]	Melting and solidification, numerical	A method for measuring the thermoelectric power of metals near their melting point and the thermo-junctions formed between them for several phase change materials (Sn, Pb, In, Bi, Cd, and Ga) has been proposed	It was shown that during melting, a thermoelectric power jump in the negative direction for Pb, In, Cd, and Ga and in the positive direction for Bi and Sn was obtained. Furthermore, premelting trends in "In" and "Cd" solids and pre-freezing trends in "Sn" and "Bi" melts, respectively, were found
[143]	Melting and solidification, experimental	The analysis of melting and freezing performances as a function of the size of indium (In) pores confined in silica glasses was carried out. In this investigation, special attention was paid to the analysis of the influence of pore size on the melting temperature and latent heat	The "In" pore melting temperature has been shown to decrease with decrease in pore size It was also shown that in the smallest pores, the latent heat was reduced compared to the bulk value
[144]	Melting and solidification, numerical and experimental	Solidification and melting processes of n-octadecane PCM encapsulated in a novel pear-shaped thermal energy storage (PS-TES) system with and without "constraint" were carried out	It has been shown that the presence of adiabatic "constraint" reduces the melting rate In addition, it was found that the complete melting time of the PS-TES system is less than that of the cylindrical case
[145]	Melting and solidification, numerical	A two-dimensional time-dependent simulation model for the melting and solidification of a PCM, filled in a system having different zigzag angle orientations and different heat transfer fluid (HTF) inlet "flow rate" and "average temperature," has been developed	PCM melting and solidification rates have been shown to increase with increase in zigzag angles In addition, the use of multiple PCM layers and increasing the HTF temperature induce faster phase change rates
[146]	Melting and solidification, numerical	The melting and solidification of PCM filled in an enclosure comprising a circular wavy heating element of various configurations and nanoparticles were considered	The wavy heater configuration has been shown to have a significant effect on controlling the convective flow characteristic within the enclosure
[147]	Melting and solidification, numerical and experimental	The melting and solidification of a phase change material filled into an enclosure with different fin surface areas were investigated	Fin surface area was found to have positive effects on conductive heat transfer, but negative effects on natural convection flow
[148]	Melting and solidification, numerical and experimental	Melting and solidification in a rectangular finned PCM tank installed radially on hot and cold-water pipes were investigated	It was found that the energy transfer efficiency is 72.4% for the tank with fins, compared to only 47.3% for a tank without fins

Table 4 (continued)

References	Process/type of analysis	Description and parametric details	Main results
[149]	Melting and solidification, numerical	A novel LHS storage system integrating two different types of conventional PCMs with different melting temperatures was proposed	A significant increase in the prediction of the behavior of thermal systems was noted. In addition, a significant reduction in production and storage time of approximately 20 min was achieved
[150]	Melting and solidification, numerical	The melting and solidification processes of a phase change composite material composed of paraffin wax (98%) and copper nanoparticles (2%) filled in a rectangular enclosure with two different geometries “deep” and “shallow” have been studied	The charging and discharging rates for the “shallow” enclosure are faster than those for the “deep” enclosure
[151]	Melting and solidification, experimental	Melting and solidification in latent heat storage (LHS) systems using paraffin wax as phase change material and with different HTF flow rates were studied	Charging and discharging times have been shown to decrease with increase in the HTF flow rate Furthermore, it has been found that the discharging process is slower than the charging process
[152]	Melting and solidification, numerical	An analytical model in which the phase change process is divided into three parts: fully solid PCM, partially molten PCM, and fully liquid PCM was developed	It has been shown that increasing the heat transfer coefficient, h , does not significantly affect PCM melting; but it affects PCM solidification. It is therefore an important parameter to control the solidification time of PCM In addition, it was found that the melting time can be increased by increasing the thickness of the PCM sample
[153]	Melting and solidification, experimental	A latent heat storage (LHS) system using microencapsulated PCM combined with carbon fibers as a latent energy storage medium has been proposed	It has been shown that the dominant mode of heat transfer is gradually transferred from an energy “storage” or “release” process to direct heat transfer between heating water and cooling water in stable states lasting around 7500 s for the initially solid PCM Furthermore, it was reported that under the same conditions, the stable temperature of the energy storage (LHS) unit is higher for the initially melted PCM
[154]	Melting and solidification, numerical and experimental	The melting and solidification of an unfixed (free) PCM, placed between two concentric horizontal isothermal cylinders, were analyzed numerically and experimentally	During the solidification process, concentric layers of PCM solid were shown to form on both walls of the tube and slowly expand into the ring. In addition, the solid phase remains in contact with the upper part of the inner tube as well as with the lower part of the outer tube On the other hand, it has been shown that heat transfer by natural convection ensures PCM melting at the top interface and that melting rates are relatively low
[155]	Melting and solidification, numerical	The melting and solidification processes of a PCM, placed in an aluminum plate-and-fin thermal storage system, with different values of inlet temperature and flow rate of the HTF fluid, were studied	It was found that the duration (time) of both phase change processes decreases with increase in HTF fluid flow rate On the other hand, melting time has been shown to decrease with increase in inlet temperature. However, the solidification time increases with increase in inlet temperature

Table 4 (continued)

References	Process/type of analysis	Description and parametric details	Main results
[156]	Melting and solidification, numerical	The solidification and melting processes of sodium nitrate (NaNO_3) as PCM placed in an LHS unit were modeled and studied numerically. The proposed model considers different values of the mushy zone constant	Charging and discharging times have been shown to increase by increasing the mushy zone constant
[157]	Melting and solidification, experimental	The solidification and melting of the two test PCMs “paraffin wax” and “stearic acid” were studied experimentally. The study considers that one side of the PCM is heated, while the opposite wall is kept constant and the other walls are insulated	It was found that the melting process starts from the upper part of the thermal storage system; while solidification takes place from below Furthermore, paraffin wax has been proven to be better than stearic acid as a means of storing thermal energy. Indeed, their measured stored thermal energies are, respectively, 61.84 kJ/kg and 57.39 kJ/kg
[158]	Melting and solidification, numerical	Melting and solidification processes of a PCM filled in an enclosure subjected to a constant and variable heat flux were studied	Melting time is shortened with variable heat flux compared to constant heat flux In addition, it was found that the melting time of PCM is shorter than the solidification time
[159]	Melting and solidification, numerical and experimental	Melting and solidification processes of different forms (spherical, cubic, triangular...) of macro-capsules containing PCM were studied	The triangular capsule was shown to exhibit better thermal efficiency with melting and solidification times of 41 and 133 min, respectively. In addition, the melting rate increases with increase in the HTF inlet temperature On the other hand, it is found that a 27% reduction in capsule size results in a reduction in melting and solidification time by 12.19 and 19.17%, respectively
[160]	Melting and solidification, numerical	Melting and solidification in a fixed-bed latent heat storage system were studied. The storage system was filled with spherical sodium nitrate (NaNO_3) capsules as PCM and used “Therminol 66” as HTF	It has been shown that the solidification time is longer than the melting time. Also, it has been shown that as the size of the PCM capsule increases, the natural convection effect in the molten PCM increases On the other hand, it was found that charging and discharging times decrease with increase in “capsule size” and “HTF fluid flow” or with decrease in Stefan number
[161]	Melting and solidification, numerical and experimental	Melting and solidification processes of epoxy resin paraffin wax as a phase change material having a melting point of 27 °C in a proposed new energy storage system were investigated	Increasing the PCM content has been shown to increase the energy storage and release capacity of the proposed system Furthermore, it was found that the experimental results agree with the numerical results
[162]	Melting and solidification, numerical	Melting and solidification in an LHS storage system with different sidewall shapes based on a “granular” phase change material and with a “gaseous” heat transfer fluid (HTF) were investigated	It was demonstrated that the straight wall shape of the studied LHS system is the key parameter to improve the charging and discharging process

Table 4 (continued)

References	Process/type of analysis	Description and parametric details	Main results
[163]	Melting and solidification, numerical	Melting and solidification in an LHS system filled with three PCMs and different HTFs: (i) HTF, (ii) solar field-HTF and (iii) thermal energy storage system-HTF (TES-HTF) were analyzed	It has been shown that in the third case of the HTF (iii) TES-HTF, an increase in its inlet temperature can increase the overall maximum energy efficiency of the system compared to the base case (i) Furthermore, it was also found that in the second case of HTF (solar field-HTF), an increase in its inlet temperature can only increase the maximum overall energy efficiency when the number of NTUs is greater than 2. In this sense, it was found that an increase in the number of NTUs can improve the maximum overall energy efficiency, provided that it does not exceed 5 On the other hand, it has been shown that increasing the number of NTUs of the HTF solar array (from case (ii)) is more effective in improving the overall maximum energy efficiency than increasing the number of NTUs of the TES-HTF case
[164]	Melting and solidification, numerical	A method for monitoring the melting (also solidification) front of phase change materials (PCM) in triplex horizontal tube heat exchangers (THI) has been developed	It was demonstrated that the developed method can represent an alternative to the effective thermal conductivity method. Indeed, the method can provide results of the order of $\pm 15\%$ compared to those provided by the effective thermal conductivity method
[165]	Melting and solidification, numerical	The melting and solidification processes in a PCM-filled thermal energy storage (TES) system with parallel spiral heat transfer fluid (HTF) pipes and under various inlet temperature conditions were investigated	Increasing the inlet temperature of HTF into spiral pipes has been shown to improve the thermal energy storage rate In addition, the energy storage (charging) time is 25% faster than the energy release (discharging) time

heat transfer becomes the most important, thus affecting the melting process [89, 90]. This analysis also shows the effect (influence) of different parameters, such as the geometric shape of the storage domain (PCM tank), the thermophysical properties of the PCM material, the operating conditions (initial and boundary conditions) imposed on the PCM domain, the orientation of the domain and the flow rate and temperature of the HTF fluid at the entrance (inlet) to the storage domain on the melting process.

Effect of the geometric shape of the storage tank on melting

The geometric shape of the domain (PCM tank) where PCM melting occurs has a significant effect on the melting process. Indeed, from the analysis carried out on the studies in Table 4 and other studies in the literature, we have realized that the following geometric shapes are the most adopted and used by different researchers and significantly influence the PCM melting process.

In a *rectangular* domain heated on one side, melting begins with a thin homogeneous layer (zone) adjacent to the heated side (wall) of the container and progresses toward the liquid–solid interface formed in a concave shape. At the beginning, the layer (melt zone) is very thin and the viscous

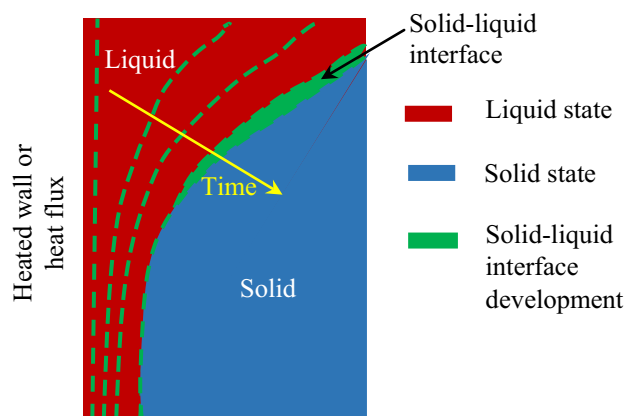
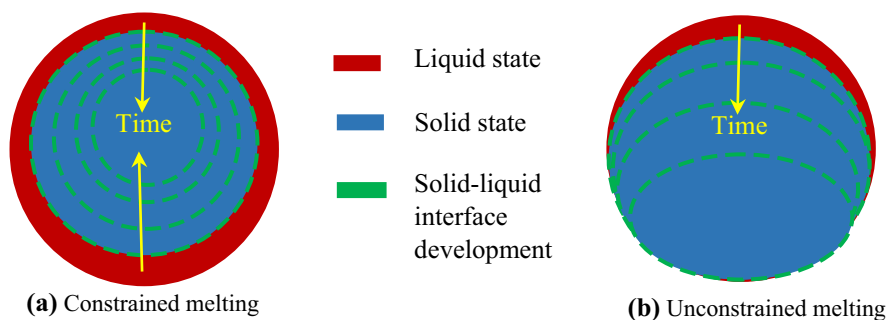


Fig. 14 Schematic representation of the evolution of the melting contour as a function of time in a PCM in a rectangular enclosure (adapted from [91]). The blue zone corresponds to the solid phase and the red to the liquid

Fig. 15 Schematic representation of the “constrained” and “unconstrained” melting processes of a PCM inside a sphere (adapted from [92])



drag forces from the walls are too high to be overcome by the effect of buoyancy forces, induced in the liquid zone due to the temperature difference (induced by density difference). As a result, the melting process is dominated by heat transfer by conduction. As the thickness of the zone increases and the viscous drag of the walls decreases, the buoyancy effect can create instability in the liquid layer; and this nonequilibrium state will cause the liquid (PCM) to rise at the start of the process. An example of studies carried out to illustrate this trend is given in the numerical results of Hamad et al. [91] presented schematically in Fig. 14.

Melting in a *spherical* domain (also called spherical shell) can be divided into two categories: constrained (fixed) melting and unconstrained (free) melting, an example study is given in reference [92] and schematized in Fig. 15. In the unconstrained case, melting is initially dominated by conduction. Over time, the molten PCM (hot liquid) rises to the top of the sphere and the “cold” solid PCM sinks to the bottom of the sphere due to the difference in density. Conversely, in the case of a constrained melting, the conduction is only significant at the beginning of the process, forming a thin layer of liquid between the wall of the sphere and the solid PCM. Over time, the liquid layer extends and expands concentrically inward due to natural convection dominating the process in the upper region and sides of the sphere. Other studies have been carried out in this context, for example, the studies of Assis et al. [93] and Sattari et al. [94] who give experimental and numerical investigations on the unconstrained and constrained melting of a phase change material in a spherical shell.

In *cylindrical* domains, melting is affected by the orientation of the axis of symmetry of the cylinder (vertical or horizontal direction). The behavior of the PCM melting process in a vertical cylinder is similar to that in the rectangular domain. Typically, the cylinder is heated from the side and/or from the base (bottom surface) or both. Melting starts from the hot cylinder wall, forming a thin layer of liquid along the wall, indicating that conduction dominates heat transfer at the start of melting. Over time, the liquid layer thickens in the upper part of the domain (cylinder) due to the development of natural convection driven by buoyancy forces, while the liquid layer across

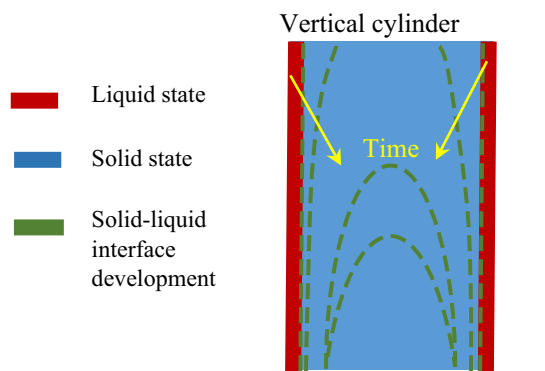


Fig. 16 Schematic representation of the evolution of the melting process as a function of time of a PCM in a vertical cylinder (adapted from [95])

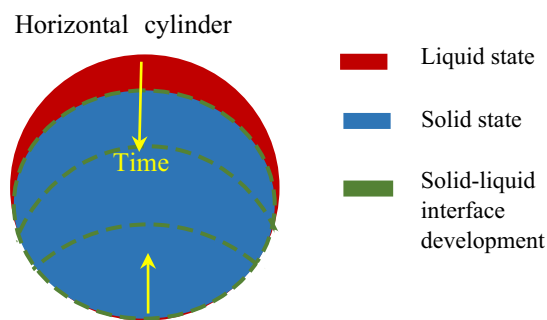


Fig. 17 Schematic representation of the temporal evolution of the melting front of a PCM in a horizontal cylinder (adapted from [96])

the entire height of the cylinder remains uniform, indicating that conduction is still significant in this region. As the influence of natural convection increases, the thickness of the liquid layer gradually increases along the cylinder. An example study illustrating this case is given by Jones et al. [95]. Indeed, through this study, the authors indicated the production of four distinct regimes which govern melting, namely: (a) the regime dominated by pure conduction, (b) the regime dominated by the combination of convection and conduction, (c) the regime dominated by convection, and (d) the regime dominated by shrinkage of the PCM solid, as shown schematically in Fig. 16. In contrast, PCM melting in a *horizontal* cylinder is similar to unconstrained melting in a spherical shell. In this case, natural convection is dominant during the melting which progresses toward the interior of the cylinder. In addition, a secondary vortex (circulation) (i.e., secondary convection cell) occurs and significantly affects the lower part of the melting region in the cylinder, as shown schematically in Fig. 17. The thickness of the melting layer (zone) decreases circumferentially from the top to the bottom of the cylinder due to convection movement; so that the molten material is carried upward along the cylinder wall

and gains (absorbs) heat. Then, the hot melt material falls (descends) along the solid–liquid interface and loses heat and this downward convective flow directly contributes to the melting process. An example study describing the melting behavior in a horizontal cylinder is given in the experimental results of Ho et al. [96].

Effect of thermophysical properties of PCM on the melting process The thermophysical properties of PCM material such as thermal conductivity, k , melting temperature (melting point), T_m , latent heat of fusion (melting), H_L , and viscosity, μ , significantly affect the melting process. Using a PCM with high thermal conductivity speeds up the melting process; conversely, using a PCM with a high melting temperature results in a longer melting time, Fan et al. [97], which guides us to choose PCMs with low melting temperature and high thermal conductivity. On the other hand, using a high latent heat value of PCM increases the melting time (i.e., speeds up the melting), Li et al. [98]. It should be added that the solid–liquid interface is also affected by the viscosity of the PCM; indeed, the melting rate is proportional to the viscosity value of the PCM, Kasibhatla et al. [99]. Note that the melting rate is defined as the energy stored in the storage system (PCM domain) divided by the melting time.

Effect of operating conditions (mainly initial and boundary conditions) on the melting process The type of thermal boundary conditions imposed on the wall of the domain (PCM tank) containing the solid (PCM) and from which melting must initiate, plays an important role in the performance of the PCM melting process. Among these types, the specified temperature condition and the convective heat flux condition are distinguished. Several studies in the literature (see also Table 4) have examined the effect of the type of thermal boundary conditions on the melting process of a PCM solid. These studies concluded that imposing a high temperature (relative to the melting point) on the wall improves the melting rate by reducing its duration (melting time), while using convective heat flux, as a boundary condition on the wall, leads more to an improvement in melting compared to the previous heated wall condition.

On the other hand, it has also been demonstrated that a high-value temperature condition (compared to the melting point) imposed on the top wall or bottom wall of the domain containing the PCM solid (PCM tank) affects its melting process. Indeed, studies carried out on this question have shown that for a domain heated from below, the melting front progresses in a planned manner and parallel to the heated wall due to the domination of conduction. In particular, in the case of a domain heated from below (bottom), where it was observed that the melting of the PCM is characterized by the appearance of an even number of rolls (Rayleigh–Bénard convection cells) at the bottom due to the

weaker effect of buoyancy forces, as shown, for example, by Duan et al. [100] and schematized in Fig. 18.

Likewise, it should also be noted that maintaining the solid PCM, in the tank before melting, at an initial temperature, $T_0 \equiv T_{in}$, having a value less than but close to its melting temperature T_m as the initial condition (i.e., at $t=0$ s) theoretically leads to a reduction in the melting time of the PCM. Nevertheless, this condition has been experimentally shown to have an insignificant effect on actual melting processes, Gao et al. [101].

Effect of orientation of the PCM tank on the process of melting The above analysis of the literature consulted as part of this work and presented in Table 4 also shows that for different orientations of the PCM tank, the melting of the PCM is characterized by the appearance of a solid–liquid interface which begins planar and progresses and advances parallel to the heated wall; then it becomes concave in shape, which enlarges in the upper part of the domain (tank of PCM solid). This curvature decreases and the melting rate slows

with increase in orientation (inclination) angle. An example of a study of this behavior is given by Zeng et al. [102] and is schematized in Fig. 19.

Effect of conditions and flow characteristics of the heat transfer fluid on the melting process The analysis of the literature consulted here and presented in Table 4 also shows that the melting process can be accelerated by using a high HTF flow rate and a high value of the HTF temperature at the inlet of the LHS unit; these make it possible to bring more heat to the PCM to absorb it during its melting.

Solidification process

On the other hand, the analysis of the studies presented in Table 4 indicates also that, unlike the melting process, in the solidification process heat is transferred initially (at the first stages of the process) by convection and later by conduction only. The solid region is formed at the heat transfer surface and acts as a resistance to heat transfer. Likewise,

Fig. 18 Schematic representation of the melting fronts developed when different cases of heated wall are applied to the PCM domain (adapted from [100])

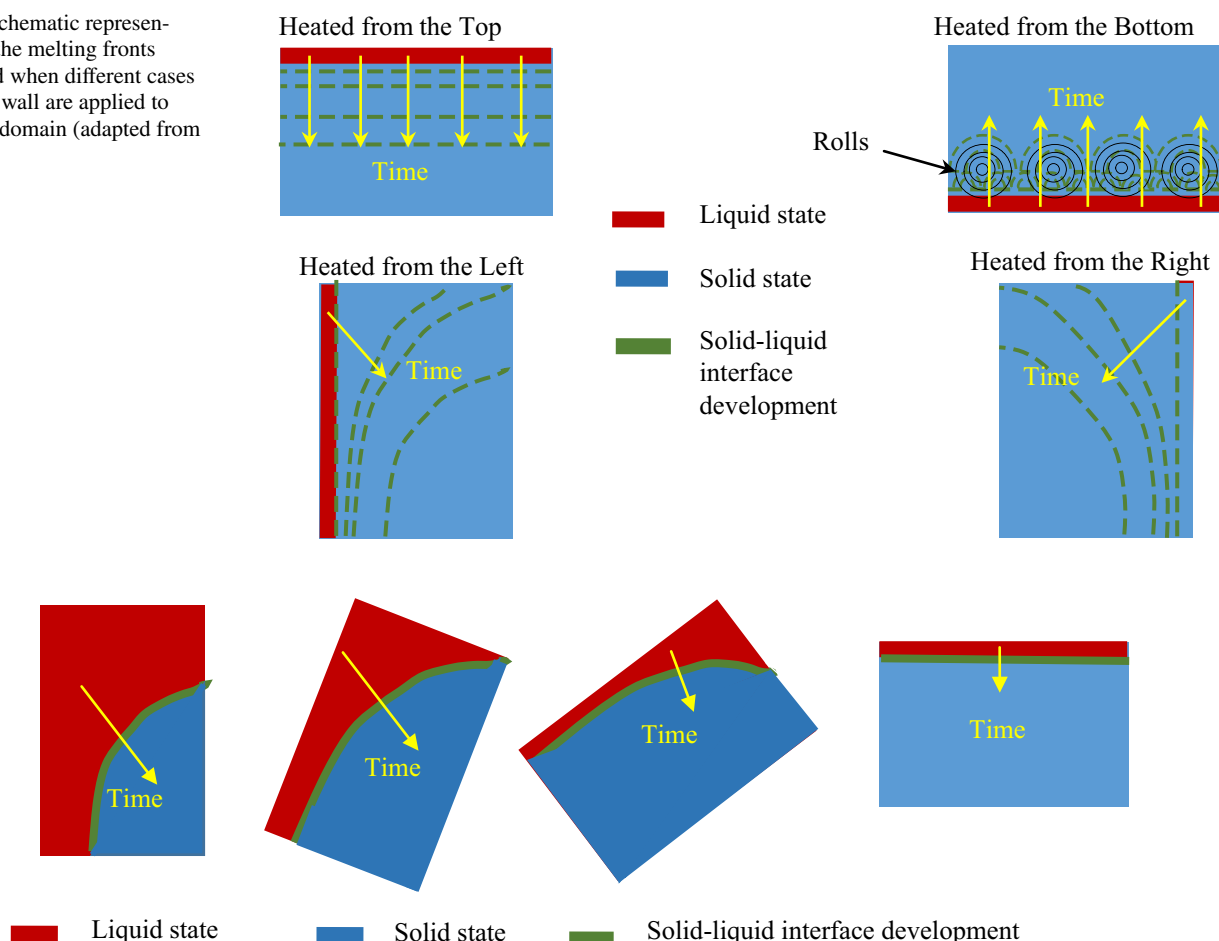


Fig. 19 Schematic representation of the evolution of the solid–liquid interface during the melting process with different domain inclination angles (adapted from [102])

the analysis also shows the influence of various parameters, such as geometry and material properties, boundary conditions and container orientation, on the solidification process. Similarly, the analysis also shows the influence of various parameters including the geometric shape of the PCM container, the properties of the PCM material, the boundary conditions imposed on the PCM container, and the orientation of the container on the solidification process.

Effect of the geometric shape of the storage domain on the solidification process Much like melting, the shape of the domain (PCM tank) in which PCM solidification occurs has an important effect on the solidification process. Similar to melting, through the bibliographic analysis of the studies presented in Table 4 we also realized that the following shapes of the domain, affecting the solidification of a PCM, have been considered by different researchers in the literature.

In a rectangular domain, solidification begins at the cold wall, and over time, the quantity of solid PCM (solid fraction, f_s) becomes greater, particularly in the lower part of the container. This phenomenon is attributed to a variation in density, whereby the solid, being denser than the liquid, sinks to the bottom. This formed solid part promotes heat transfer by conduction and at the same time acts as thermal resistance between the “cold” solid PCM and the “hot” liquid PCM. This dual role therefore hinders the solidification process. A relevant example of solidification (freezing) of a PCM in a rectangular domain (enclosure) is the analytical and numerical study by Darzi et al. [103], as shown schematically in Fig. 20.

Among existing PCM domains, spherical shells perform the best, due to their high external surface area relative to the amount of PCM that can be encapsulated [104]. In this configuration, the PCM starts to solidify from the

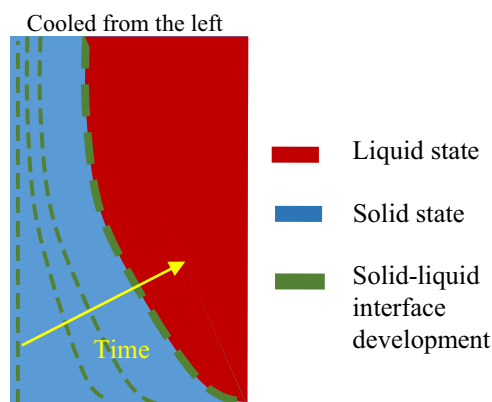


Fig.20 Schematic representation of the temporal evolution of the PCM solidification front inside a rectangular domain. The blue region corresponds to the solid phase (solid fraction) and the red to the liquid (liquid fraction, f_l) (adapted from [103])

cold outer wall and moves inward toward the center of the sphere, forming a significant solid part in the lower area due to natural convection, as shown in Fig. 21. The densest solid PCM is located at the lower region (bottom) of the cavity; so buoyancy forces are neglected. This phenomenon has been explained in more detail, for example, in the numerical studies of Ghosh et al. [105] and Asker et al. [106].

In addition to spherical shells, cylindrical domains are also favored for thermal energy storage applications due to their optimal surface-to-volume ratio [107]. In a vertical cylinder, the solidification front is similar to that of a rectangular domain. At the beginning of the process, heat transfer is dominated by natural convection, driven by buoyancy forces, so that the “hot” liquid PCM is pushed to the top of the cylinder while the “cold” liquid occupies the part bottom of the cylinder. Over time, solidification begins inward, forming a solid layer along the length of the cylinder and the mode of heat transfer shifts to conduction dominance. As solidification progresses a significant portion of solid PCM forms at the bottom of the cylinder. As an example of numerical results, we cite the study of Izgi et al. [108] where the solidification of a PCM inside a vertical cylinder was discussed and presented, as schematically presented in Fig. 22. While in the case of a horizontal cylinder, the development of the solid–liquid interface (front) is similar to that of the sphere. Solidification begins from the internal wall of the cylinder and progresses radially toward the center, particularly toward the lower zones due to natural convection. This mode of heat transfer dominates in the upper zones of the cylinder because the buoyancy forces direct the “hot” PCM liquid upwards and the “cold” liquid downwards, thus promoting solidification in the lower part, where conduction is dominant. This phenomenon is described in more detail, for example, in the numerical results of Mozafari et al. [109] as shown schematically in Fig. 23.

Effect of material properties on the process of solidification During the freezing process, solidification is affected by the thermal conductivity, latent heat, and melting point

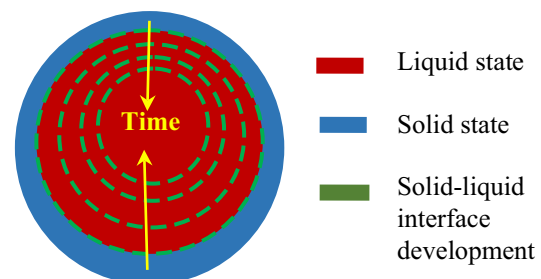


Fig.21 Schematic representation of the temporal evolution of the solid–liquid interface during solidification of PCM in the sphere (adapted from [105])

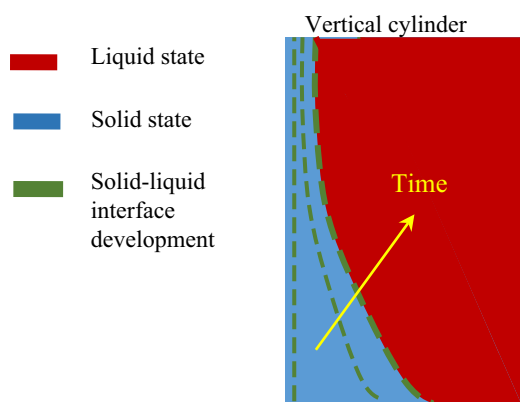


Fig. 22 Solidification process inside a vertical cylinder (adapted from [108])

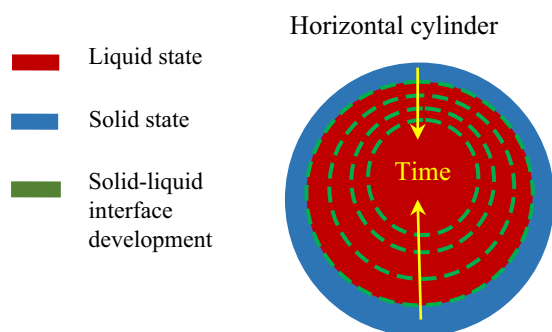


Fig. 23 Schematic representation of the temporal evolution of solidification inside a horizontal cylinder (adapted from [109])

of PCM. Most of the studies reviewed in this work (such as those presented in Table 4 and others) have shown that increasing the thermal conductivity of PCM improves the solidification rate by decreasing its duration (solidification time), for example, the study by Panchabikesan et al. [110]. In addition, the longest solidification time (implying a slow process) is obtained for PCM with a low melting point, Fan et al. [96]. These trends guide us in choosing PCMs with low melting points and high thermal conductivity to improve solidification.

Effect of operating conditions (mainly initial and boundary conditions) on solidification The thermal boundary condition imposed on the wall of the PCM tank to begin the solidification process has a significant impact on this process. A review of the studies examined here showed that the rate of solidification is inversely proportional to the temperature or heat flux imposed on the wall of the PCM tank to initiate the process; it decreases by increasing the temperature or the flux on the wall. Likewise, the solidification rate is inversely proportional to the initial temperature imposed on the liquid PCM, Gao et al. [100].

Effect of orientation of the PCM tank on the process of solidification Unlike the melting process, solidification is accelerated by increasing the inclination angle of the PCM tank, Iachachene [111].

Effect of conditions and flow characteristics of the heat transfer fluid on the solidification Similarly, a review of the literature consulted here and presented in Table 4 shows that the solidification process can be accelerated by increasing the flow rate of the HTF, which in turn improves the heat transfer rate between the HTF and the PCM. However, the solidification rate decreases when the temperature of the HTF at the inlet of the LHS unit is higher.

Thermal performance parameters for a PCM performing a phase transition

A PCM can be characterized by its thermophysical properties, including its thermal conductivity, temperature of phase change, and latent heat of phase change. However, it should be noted that it is also characterized by its ability to achieve a phase transition in a short period (short transition time, t_m), with a large amount of total energy stored, E , and with a high rate of transmitted heat (heat transfer rate, $\dot{q}_m \equiv \dot{q}_{PCM}$). Indeed, these latter parameters are essential and represent the main *criteria* for measuring the heat transfer performance of a PCM performing a phase transition (Fig. 24) and therefore of its choice for a particular application. In what follows, particular attention will be paid to these parameters.

Remark 2 For the sake of simplicity without loss of generality, in the following subsections, attention is paid only to the heat transfer performance parameters of the melting process.

Duration time or time for storage (melting time)

The melting process duration of a particular PCM refers to the time required to transition from its solid phase to its liquid phase during the process. It depends on the shape, size, or volume of the PCM domain, the thermal conditions imposed on the domain, and the amount of PCM in the domain [166]. To estimate the melting duration (melting time), many studies have been carried out. These studies made it possible to develop correlations based on the Fourier number (Fo) as a function of the Stefan number (Ste) and (i) the Rayleigh number (Ra), (ii) the Reynolds number (Re), or (iii) the numbers Ra and Re, depending on the type of heat transfer that dominates the transition (melting) process (i.e., conduction, natural convection, forced convection, or mixed convection), as indicated in Table 5.

As an example without loss of generality, for a cylinder filled with a PCM (domain where the PCM melting process takes place) shown in Fig. 24c. The Fourier number (Fo) in

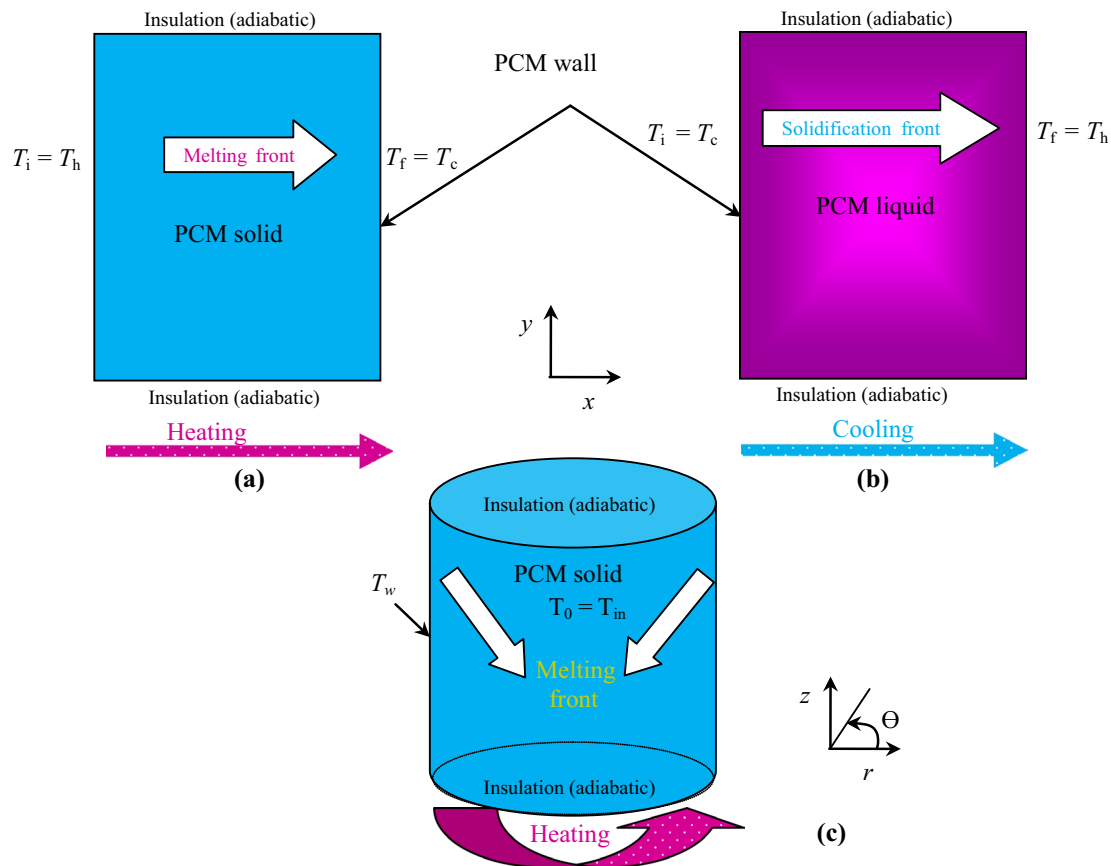


Fig. 24 PCM performing phase change in different physical domains. Calculation of PCM heat transfer performance parameters

Table 5 Criteria for choosing melting time correlation (case of PCM melting process in a cylinder)

References	Geometry	Restrictions	Melting time correlation	Validity & comment
[167]	Cylinder and slab	$0^{\circ}\text{C} \leq T_m \leq 70^{\circ}\text{C}$ $0 < \text{Ste} \leq 4$	$\text{Fo} = 0.11 + 0.25/\text{Ste}$ (for cylinder) $\text{Fo} = 0.125 + 0.5/\text{Ste}$ (for slab)	Conduction-dominated melting
[168]	Vertical cylinder	$0.08 \leq \text{Ste} \leq 0.325$	$\text{Fo} = 0.25 + 0.134/\text{Ste}$	Conduction-dominated melting
[169]	Horizontal cylinder	$0.2 \leq \text{Ste} \leq 0.357$	$\text{Fo} = 0.14 + 0.15/\text{Ste}$	Conduction-dominated melting
[170]	Symmetric cylinder	$0.02 \leq \text{Ste} \leq 0.5$ and $T_m = T_{in}$	$\text{Fo} = 0.14 + 0.25/\text{Ste}$	Conduction-dominated melting
[171]	Vertical cylinder	NA (not available)	$\text{Fo} = 0.515 \cdot (\text{Ste}^{-1.24} \text{Re}^{-0.133})$	Forced convection dominated melting
[172]	Vertical cylinder	$3.04 \times 10^5 \leq \text{Ra} \leq 65.75 \times 10^5$ $0.2 \leq \text{Ste} \leq 1$ $600 \leq \text{Re} \leq 3000$ $2 \leq L/D \leq 15$	$\text{Fo} = \frac{12.5587(L/D)^{0.3019}}{\text{Ste}^{0.8355} \text{Ra}^{0.1289} \text{Re}^{0.0892}}$ with L/D is cylinder length to diameter ratio	Mixed convection dominated melting
[173]	Vertical cylinder	$3500 \leq \text{Re} \leq 8500$ $0.3 \leq \text{Ste} \leq 0.65$ $0.3 \leq \theta_T \leq 0.4$	With θ_T is a temperature constant: $\theta_T = T_{in}/T_w$	Forced convection dominated melting

terms of melting time (t_m), as well as the Stefan number (Ste) are, respectively, given as follows:

$$\text{Fo} = \frac{\alpha_l t_m}{R^2}$$

(1)

$$\text{Ste} = \frac{C_{pl}(T_w - T_m)}{H_L} \quad (2)$$

where in Eq. (1) α_l is the thermal diffusivity of the PCM liquid, t_m is the melting time (with $t_m = R^2 \text{Fo} / \alpha_l$) and R is the radius of the cylinder (PCM domain). In Eq. (2) C_{pl} is the

PCM liquid-specific heat, T_w is the wall temperature of the cylinder, T_m is the melting point of PCM and H_L is the PCM latent heat of phase transition (melting). Note that the thermal diffusivity of the PCM melt is given by $\alpha_l = k_l / \rho_l C_{pl}$, where ρ_l is the density of the PCM melt.

If melting is dominated by free convection, we should introduce Ra as

$$Ra = \frac{g\beta(T_w - T_m)R^3}{\alpha_l \nu} \quad (3)$$

where g is the gravity ($g = 9.81 \text{ m/s}^2$), β is the thermal expansion coefficient of the PCM liquid, and ν is the kinematic viscosity of the PCM liquid.

If melting is dominated by forced convection, we should introduce Re as

$$Re = U_{ref} D / \nu \quad (4)$$

where U_{ref} is the reference velocity and ($D = 2R$) is the diameter of the cylinder (PCM melting domain).

Energy stored (amount of storage energy)

In general, energy stored (heat stored) is the amount of energy absorbed by the PCM when it melts or released by the PCM when it solidifies as latent and sensible heat. To quantify the total energy stored (or total amount of storage energy) in the PCM during its melting, for example, three stages must be considered: (1) before, (2) during and (3) after the melting of the PCM, such as:

$$E = E_{S,i} + E_{L,m} + E_{S,f} \quad (5)$$

where the subscripts i , m and f , respectively, refer to the stages “before,” “during,” and “after” the phase change, and E is the total energy stored in the PCM, $E_{S,i}$, $E_{S,f}$ are the sensible heats stored before and after the phase change, respectively, and $E_{L,m}$ is the latent heat stored during the phase change.

Energy stored by sensible heat (sensible heat stored) before and after the melting process The total energy stored by the sensible heat results from the rise in the temperature of the PCM during its heating from an initial temperature T_i to a final temperature T_f ; (see Fig. 24a) it is defined as follows:

$$E_S = E_{S,i} + E_{S,f} = \int_{T_i}^{T_m} m C_{p_s} dT + \int_{T_m}^{T_f} m C_{p_l} dT = m(C_{p_s}(T_m - T_i) + C_{p_l}(T_f - T_m)) \quad (6)$$

where m is the mass of PCM to be heated in its physical domain (e.g., Figure 24a), C_{p_s} , C_{p_l} are the PCM specific heat at the solid and liquid states, respectively. T_i , T_m , and T_f are the initial, melting, and final temperatures, respectively.

Energy stored by latent heat (latent heat stored) during melting process The energy stored by latent heat (i.e., the latent heat stored by the PCM during its phase change at T_m) is derived from the phase change of the PCM from a solid state to a liquid state; we can write it like this:

$$E_{L,m} = m_{s \rightarrow l} H_L \quad (7)$$

where $m_{s \rightarrow l}$ is the “molten” mass of the PCM in the domain (e.g., Figure 24a). Note that $m_{s \rightarrow l} = m f_l$, with the fraction of PCM liquid, f_l , is generally taken 1 at the end of melting).

Total energy stored by PCM during its melting process By substituting Eqs. (7) and (6) into Eq. (5), we can have the total energy stored by the PCM during its melting process

$$E = E_{S,i} + E_{L,m} + E_{S,f} = m(C_{p_s}(T_m - T_i) + C_{p_l}(T_f - T_m)) + m H_L \quad (8)$$

Volumetric heat storage capacity (volumetric energy stored density) of the PCM The volumetric heat storage capacity of the PCM domain is the total heat stored by the PCM divided by the product of the storage time and the volume of the domain (PCM tank):

$$E_V = E / (t_m V_{\text{tank}}) \quad (9)$$

In the case where the PCM tank is a cylinder shown in Fig. 24c; so one can have: $E_V = 4E / \pi D^2 L t_m$. Equation (9) clearly indicates that storing thermal energy in storage systems using PCMs is a very efficient storage technique; it indeed has the advantage of having an increased heat storage capacity when the volume of the storage system is small.

Heat transfer rate (melting rate)

The heat transfer rate to or from the PCM, respectively, refers to the amount of heat absorbed (stored) or released by the PCM per unit time [174, 175]. For example, for the melting process, the amount of heat stored by the PCM per unit time (i.e., the melting rate) is:

$$\dot{q}_m = \frac{E}{t_m} \quad (10)$$

Note that the melting rate $\dot{q}_m \equiv \dot{q}_{\text{PCM}}$ is sometimes also called the average storage rate [176].

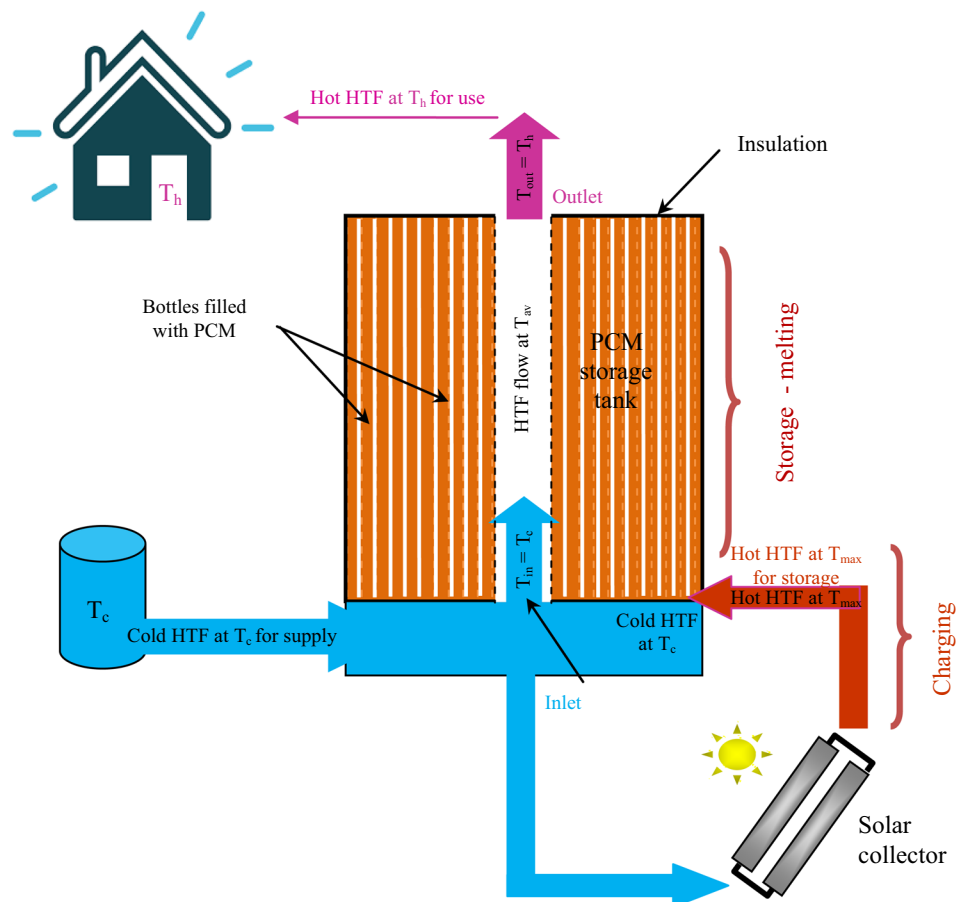
Thermal performance parameters of a PCM-based TES system

In the preceding section, we highlighted the heat transfer performance parameters relating to a PCM (included in a heat storage domain) undertaking a phase transition. If in addition, the domain of the PCM (PCM tank) is crossed by a flow of heat transfer fluid (HTF), it results that a thermal exchange between the PCM and the HTF can occur. This is because if the HTF passing through the PCM tank through

the internal channel is hot (coming from a hot heat input source); it allows the PCM to melt and the heat is stored inside the PCM tank (charging process (cycle)). Whereas if the HTF, passing through the PCM tank, is cold (coming from a cold heat supply source); it allows the PCM to solidify, and the heat is released outside the PCM tank (discharging cycle). In this scenario, the situation no longer corresponds to a PCM storage domain undertaking a phase transition but to that of a PCM-based TES system (often referred to as an LHS unit (Fig. 25)). In this scenario, the situation no longer corresponds to a storage domain where the PCM undertakes a phase transition but to that of a PCM-based TES system [177, 178] (often called an LHS unit (Fig. 25)).

The main characteristics of an LHS storage unit quantifying their thermal performance and which are taken into account in its selection in a particular application are generally summarized as energy stored per unit of time (heat transfer rate), total energy stored, and time for storage/mass of hot HTF prepared (in the PCM storage tank) for use [177]. The LHS storage unit is qualified as efficient if it combines the following elements: high and longer energy storage capacity, fast charging (sufficiently short storage time), low-temperature fluctuation of the heat transfer fluid at the HTF channel output (production) during discharge,

Fig. 25 Thermal performance parameters of a PCM-based heat storage system (inspired from [177])



and uniform charge/discharge rate [178]. Similarly, for the sake of simplicity without loss of generality, we should focus only on the heat transfer performance parameters of the melting process.

Energy stored per time unit (heat transfer rate)

If we consider the energy storage system like that shown, for example, in Fig. 22, the energy stored per unit of time (or heat transfer rate)[177, 179] in this LHS system, is given by:

$$\dot{Q}_{\text{HTF}} = \dot{m}_{\text{HTF}} C_{p,\text{HTF}} (T_{\text{out,HTF}} - T_{\text{in,HTF}}) \quad (11)$$

where $C_{p,\text{HTF}}$ is the specific heat of the HTF passing through the storage system, $\dot{m}_{\text{HTF}}$ is the “required” mass flow rate of the HTF passing through the storage system (through the inner channel), $T_{\text{in,HTF}}$ is the temperature of the cold HTF fluid, which arrives at the inlet of the storage system channel (i.e., initial temperature of the cold HTF at the inlet of the tank (which called also supply temperature of the storage system with the HTF fluid (initial temperature of the charging), and $T_{\text{out,HTF}}$ is the temperature of the hot HTF fluid at the outlet of the storage system channel (storage tank) “desired” for the use (e.g., domestic heating).

Total energy stored for the PCM storage tank (energy storage capacity of the LHS unit)

The principle of the LHS unit, schematized in Fig. 25, is that the system can store the heat resulting from the contribution of the increase (rise) in temperature beyond the standard conditions of the HTF fluid and the PCM (i.e., sensible heats) and the energy absorbed in the PCM due to its melting (i.e., latent heat). As a result, the total stored energy (energy storage capacity) of the LHS unit mainly depends on (i) the mass and heat capacity of the HTF (i.e., m_{HTF} and $C_{p,\text{HTF}}$) and (ii) the mass and heat capacity of the PCM (i.e., m_{PCM} and $C_{p,\text{PCM}}$ in solid and liquid phases). And it is equal to the sum of sensible heat stored and latent heat stored [107, 177, 178]. Indeed, on the one hand, the total sensible heat stored is made up of the following contributions:

$$\begin{aligned} E_{\text{S,LHS}} &= E_{\text{S,HTF}} + E_{\text{S,i pcm}} + E_{\text{S,f pcm}} = \int_{T_{\text{av,HTF}}}^{T_{\text{max,HTF}}} m_{\text{HTF}} C_{p,\text{HTF}} dT + \int_{T_i}^{T_m} m_{\text{PCM}} C_{p,\text{PCM}} dT + \int_{T_m}^{T_f} m_{\text{PCM}} C_{p,\text{PCM}} dT \\ &= m_{\text{HTF}} C_{p,\text{HTF}} (T_{\text{max,HTF}} - T_{\text{av,HTF}}) + m_{\text{PCM}} (C_{p,\text{PCM}} (T_m - T_i) + C_{p,\text{PCM}} (T_f - T_m)) \end{aligned} \quad (12)$$

where the sensible energy stored by the HTF is taken as: $E_{\text{S,HTF}} = m_{\text{HTF}} C_{p,\text{HTF}} (T_{\text{max,HTF}} - T_{\text{av,HTF}})$; here m_{HTF} and $C_{p,\text{HTF}}$ are the required mass of the HTF fluid and the specific heat capacity of the HTF fluid in the heat storage

system, respectively. $T_{\text{max,HTF}}$ and $T_{\text{av,HTF}}$ are the maximum temperature in the storage system (sensible heat storage temperature) and the average (mean) temperature of the HTF fluid flow in the channel (which can be approximated by $T_{\text{av,HTF}} = (T_{\text{max,HTF}} + T_{\text{in,HTF}})/2$), respectively.

On the other hand, the latent heat stored in the PCM is given by above Eq. (7), which can be written in this case as:

$$E_{\text{L,mLHS}} = (m_{\text{s-l}} H_L)_{\text{pcm}} = m_{\text{s-l,pcm}} H_L = m_{\text{PCM}} f_l H_L \quad (13)$$

Finally, the energy storage capacity of the LHS unit in Fig. 22 is written as:

$$\begin{aligned} E_{\text{LHS}} &= E_{\text{S,HTF}} + E_{\text{S,i pcm}} + E_{\text{S,f pcm}} + E_{\text{L,mLHS}} = m_{\text{HTF}} C_{p,\text{HTF}} (T_{\text{max,HTF}} - T_{\text{av,HTF}}) \\ &\quad + m_{\text{PCM}} (C_{p,\text{PCM}} (T_m - T_i) + C_{p,\text{PCM}} (T_f - T_m)) + m_{\text{PCM}} f_l H_L \end{aligned} \quad (14a)$$

Which can be written as:

$$E_{\text{LHS}} = m_{\text{HTF}} C_{p,\text{HTF}} \Delta T_{\text{HTF}} + m_{\text{PCM}} (C_{p,\text{PCM}} \Delta T_{i \rightarrow m} + C_{p,\text{PCM}} \Delta T_{m \rightarrow f}) + m_{\text{PCM}} f_l H_L \quad (14b)$$

In previous Eqs. (12) to (14) m_{PCM} represents the mass of the PCM in the tank, f_l is the PCM liquid (melt) fraction (which should be assumed to be 1 at the end of the charging process), $C_{p,\text{PCM}}$ and $C_{p,\text{PCM}}$ are the specific heat (heat capacity) of PCM in the solid and liquid phases, respectively. $\Delta T_{\text{HTF}} = (T_{\text{max,HTF}} - T_{\text{in,HTF}})$ is the temperature variation “required” to heat the mass of HTF in the storage tank. While the temperature rise of the PCM, composed of the contributions of: (1) $\Delta T_{i \rightarrow m} = (T_m - T_i)$ Which represents the temperature variation between the PCM melting temperature and the initial temperature of the cold HTF (at the inlet) of the PCM storage tank, and $\Delta T_{m \rightarrow f} = (T_f - T_m)$ which represents the temperature difference between the hot HTF (at the outlet) of the storage tank and the PCM melting temperature [177, 178]. For simplicity, we can assign the values of $T_{\text{in,HTF}}$ and $T_{\text{out,HTF}}$

to the “initial” temperature and the “final” temperature of the storage medium (PCM) $T_i \equiv T_c$ and $T_f \equiv T_h$, respectively; so $T_i = T_{\text{in,HTF}}$ and $T_f = T_{\text{out,HTF}}$.

Time for storage and mass of hot HTF in the PCM storage tank

Other performance parameters include time for storage and mass of hot HTF, which is “prepared” for use via the PCM heat storage tank [177]. These parameters can be expressed, respectively, by:

$$t_{\text{HTF,Use}} = \frac{E_{\text{LHS}}}{\dot{q}_{\text{HTF}}} \quad (15)$$

$$m_{\text{HTF,Use}} = \dot{m}_{\text{HTF}} t_{\text{HTF,Use}} \quad (16)$$

In the case of an LHS unit, the heat transfer rate $\dot{q}_{\text{HTF}}$ is also called the average storage rate of the LHS unit (see, e.g., [176]).

An analysis of the thermal performance of both a PCM undertaking a phase transition and an LHS unit studied, respectively, in Sects. 3.3 and 3.4 allows us to deduce that the thermal performance of PCM-based storage systems strongly depends (as already indicated in the Introduction section) on the thermal properties of the PCM, including thermal conductivity, thermal diffusivity, and heat capacity of the PCM. Improving the thermal performance of PCM is therefore strongly required. This issue should be addressed in the next section.

Heat transfer enhancement techniques

As stated before, the use of PCM in thermal storage systems offers many advantages, such as high storage capacity, excellent heat transfer performance, and near-constant operating temperature. However, despite these advantages, it should be noted that the main disadvantage of PCMs is their low thermal conductivity, thermal diffusivity, and heat capacity, which increases the storage time and reduces the energy storage efficiency (i.e., stored energy) in the storage system. Consequently, to remedy these drawbacks and thus improve heat transfer during phase change, improvement methods have been recommended and used by different researchers as shown in Table 6; these methods include adding different types of fins to the geometry of the storage unit (mainly to the PCM tank), combining the PCM with one or more highly conductive metal foams, dispersing highly thermally conductive nanoparticles in the PCM, and the application of external force fields to the system containing the PCM. These techniques can be applied alone or in combination; for details, see Table 6. It should be noted that the nanoparticles or metal foams should preferably contain high carbon nanoparticles (a considerable amount of carbon) to improve the thermal conductivity of the resulting “nano-enhanced PCM” composite.

Enhancement by incorporating fins into the PCM storage system

Principle and discussion

Improving heat transfer by incorporating (integrating) fins, on the heating (or cooling) side of the PCM in the storage tank, allows for better heat transfer to (or from) the thermal storage medium (PCM) during its intended phase transition. The integration of the fins depends on several parameters such as their length and width, their number and their position in the storage system. An example of fins integration, on the heating (or cooling) side of the PCM in a storage system, is shown schematically in Fig. 26.

Through literature analysis (e.g., in Table 6), we find that increasing the number, length and width of fins accelerates both the melting process and the solidification process.

On the other hand, to simplify without losing generality, we limit the discussion to the enhancement effect of fins on the melting process. Indeed, the literature analysis showed also that as the fin or fins, horizontally installed to the PCM tank, move away from the upper wall of the storage system, the melting process slows down; which is a disadvantage, so it is recommended to install fins close to the top of the PCM. Furthermore, a fin or fins inclined toward the top of the PCM constitute the best solution to improve its melting. Examples illustrating such trends are given by experimental and numerical results due to Zhang et al. [180] and Khatra et al. [181], respectively.

Enhancement by combining metal foams with PCM

Principle and discussion

The combination of metal foams with PCM is one of the most applied and effective techniques to improve the thermal conductivity and overall heat transfer of the resulting mixture (foam-doped PCM). Indeed, the integration of metal foam promotes the predominance of heat transfer by conduction during the melting process [182]. Generally, copper, aluminum, and nickel are the most common metals used to make metal foams. However due to its high thermal conductivity, copper is more adopted than other metals. However, in high-temperature PCM applications, low thermal conductivity nickel is most commonly used due to its higher melting temperature than copper. The use of aluminum which has a low density and low melting temperature compared to that of nickel, is limited in high-temperature PCM applications [183]. A scheme of combining metal foam with PCM, for example, of an energy storage unit is shown in Fig. 27.

In fact, the integration (combination) of metal foam into PCM (base material) is affected by various parameters such as porosity, pore size, and the nature of the base material.

Table 6 Literature review of the different techniques (ways) for improving PCM

References	Method	Process/type of study	Description and parametric details	Main results
[189]	Fin	Melting, numerical	The improvement of the melting of lauric acid (PCM), filled in an enclosure with rectangular fins, was studied by optimizing the number and inclination of the aluminum fins (2 fins and three fin inclinations from the center of the enclosure at 60°, 45°, and 35° were tested)	PCM has been shown to melt faster with a 43% reduction in melting time using two fins compared to the no-fin model. Furthermore, the 45° and 60° inclinations can reduce the melting time by 31% and 5.12%, respectively, compared to the 90° inclination
[190]	Fin	Melting, numerical and experimental	Improving the melting of a PCM in an enclosure by including aluminum fins and filling spirals in the enclosure was studied	Finned and spiral aluminum fillers have been shown to improve the PCM melting process. Additionally, adding aluminum spiral fillers allows for faster melting
[191]	Fin	Melting, experimental	The improvement of the melting of a PCM in a transparent rectangular enclosure subjected to different temperatures at its heated wall was studied using partial horizontal fins	Adding 1 and 3 fins has been shown to reduce melting time by 18% and 37%, respectively, compared to that predicted without fins On the other hand, it was found that increasing the wall temperature decreases the total melting time
[192]	Fin	Melting, numerical	The melting enhancement of a PCM in a helical fin LHS unit was investigated using different orientations, directions of these fins and different HTF fluid inlet velocities	The integration of “double helical fin” and “quadruple helical fin” has been shown to reduce total melting time by 31.0% and 10.0%, respectively, compared to conventional fin shapes. Furthermore, it has been suggested that the annular fin and the helical fin should be arranged in a vertical orientation; while the quadruple fin and the longitudinal fin must be arranged in horizontal orientation On the other hand, it has been shown that when the direction of the input HTF is opposite to the direction of gravitational acceleration, the heat transfer performance is better
[193]	Fin	Melting, numerical and experimental	The enhancement of stearic acid (PCM) melting in horizontal latent heat storage (LHS) units was studied using different fin orientations and HTF inlet temperatures	The “Y-fin” configuration was found to reduce melting time by 50.7% and increase the overall energy efficiency of the LHS unit by 10% compared to the “Y-shaped” configuration. Additionally, increasing the thermal diffusivity of the shell (of the storage unit) increased the heat storage rate by 32.40% On the other hand, it was reported that increasing the HTF temperature by 7.6% improved the system's melting performance by 80%

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[194]	Fin	Melting, numerical	Improving the melting of a PCM in a latent heat storage (LHS) unit using tree-shaped fractal (fractal tree-like) fins was considered	Optimal tree-like fins reduce the total melting time and the total storage rate of the LHS unit by 26.7% and 45.4%, respectively, compared to the case of plate fins The “optimal” tree-shaped fractal fins were shown to reduce the total melting time and total storage rate of the LHS unit by 26.7% and 45.4%, respectively, compared to the case of plate-shaped fins
[195]	Fin	Solidification, numerical	The improvement of the solidification of a PCM filled in twisted fins in a triple tube heat exchanger (TTHX) was considered	It was found that the use of twisted fins reduced the solidification time by 12.7% and 22.9% compared to the straight fin case and the non-fin case, respectively. Furthermore, increasing the number of fins from two to four and six decreases the solidification time by 11.9% and 25.6%, respectively On the other hand, it was shown that the solidification rate was reduced by 28% and 40% by increasing the HTF inlet temperature from 20 to 15 and 10 °C, respectively. Additionally, the solidification time decreases by 5% and 12% with increase in Reynolds number HTF from 500 to 1000 and 1500, respectively The solidification time can be reduced by 30.98% by using new triangular fins compared to traditional ones The solidification time is reduced with increase in length fin, decreasing initial temperature or increasing thermal conductivity The total solidification time was shortened by 13.61% when the fins were all joined to the external tube compared to that joined to the internal tube
[196]	Fin	Solidification, Numerical	Improving the solidification of a PCM inside a triplex tube thermal energy storage system using a longitudinal triangular fin was considered	
[197]	Fin	Melting & solidification, experimental	Improving erythritol (PCM) melting and solidification processes in a concentric annular storage system using longitudinal fins was considered	30 kg/min and 140 °C are the appropriate HTF mass flow rate and inlet temperature for the charging process In addition, it was found that 200 min corresponds to a discharge time of 70.9% of the total energy charged for 20 kg of erythritol It should be emphasized here that the authors studied the performance of the finned system without comparing the finned and finless systems

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[198]	Fin	Melting & solidification, numerical	Improving the melting and solidification processes of a PCM filled in a triplex tube heat exchanger (TTHX) using longitudinal fins was considered	It was found that the effect of fin thickness was insignificant, while their number and length favored heat transfer
[199]	Foam	Melting, numerical and experimental	Melting enhancement by incorporating aluminum foams into PCM paraffin wax was investigated	It was found that the incorporation of metal foam into PCM improves the melting rate by 78% compared to the case of pure PCM (without metal foam)
[200]	Foam	Solidification, experimental	The improvement of the solidification of a composite of "PCM and metal foam" by considering different parameters characterizing copper and aluminum metal foams was considered	Adding metal foams (inserts) to PCM has been shown to reduce solidification time by more than 25% On the other hand, it was found that the composition insert (40 PPI, 93.3%, copper foam) is the most efficient by improving the solidification process by approximately 28% compared to the composition insert (10 PPI, 95.5%, aluminum foam)
[201]	Foam	Solidification, experimental	The solidification enhancement of stearic acid PCM using carbon additives (multi-walled carbon nanotubes (MWCNTs), graphite and graphene) and polyvinylpyrrolidone (PVP) as dispersion stabilizer was examined	The solidification rate was shown to be improved with increase in concentrations of MWCNT, graphite, and graphene Additionally, graphite was found to be the best agent to enhance the solidification process; in fact, it increases the heat transfer rate by more than 3.35 times by adding 5 volume percent (5 vol %) graphite On the other hand, it was found that PCM with PVP gives better thermal conductivity than without PVP
[202]	Foam	Melting & Solidification, experimental	Improving the melting and solidification of PCM encapsulated in a vertical cylinder by a heat pipe (HP) located concentrically and having different configurations "HP-foil-PCM," "HP-foam-PCM," "HP-PCM" and "Rod-PCM" was examined	It was found that the performance of the "HP-foil-PCM" configuration (with a porosity of 0.957) exceeds the performance of the "HP-foam-PCM," "HP-PCM" and "Rod-PCM" configurations by 0.5, 10 and 15 times, respectively

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[203]	Foam	Melting & Solidification, numerical and experimental	The improvement of melting and solidification of “water (PCM)” and “copper (foam)” composite was studied	It was found that the total solidification and melting times of the water-copper composite (having porosities of 0.93 and 0.97) are reduced by 87.5% and 76.7%, respectively, compared to the time of solidification of pure water Moreover, the total solidification time for the composite (having the porosities of 0.93 and 0.97) and under the action of natural convection was, respectively, 85.7% and 75.8% of that without the action natural convection. Remember that porosity is a dimensionless quantity which represents the ratio between the volume of pores and the total volume of a sample
[204]	Foam	Melting & Solidification, experimental	The enhancement of melting and solidification of PCM in cylindrical enclosures using exfoliated graphene nano-additives (xGnP) was examined	It was found that the use of xGnPs additives (inserts) with concentrations of 3.0 and 4.5 mass% improves the conductivity by a factor of 2 to 3 and reduces the total times of melting and solidification up to 66% and 55%, respectively, compared to pure PCM
[205]	Foam	Melting & Solidification, numerical	Improving the melting and solidification of PCM composed of paraffin and sodium carbonate (NaCO_3) and filled in a cold energy storage system, using the incorporation of foam aluminum (Al) was studied	It was pointed out that the incorporation of aluminum foams into PCM improved the “effective thermal diffusivity” and “effective thermal conductivity” by more than 21 times and increased the discharging rate by about eight times compared to the case of PCM alone On the other hand, the cold discharging rate of NaCO_3 is higher than that of paraffin In addition, the cold discharging rate of paraffin-Al is higher than that of NaCO_3 -Al. Finally, the effective thermal diffusivity of paraffin-Al was found to be 42.5% higher than that of NaCO_3 -Al Increasing PCM mass and nanoparticle fraction has been shown to decrease melting time
[206]	Nanoparticles	Melting, numerical	Improving the melting of a phase change material in an enclosure by integrating copper nanoparticles into the PCM was studied	Increasing the charge of CNT has been shown to speed up the melting process. Furthermore, increasing the charge of CNT also had a negative effect on the melting rate at high values of the Grashof number Gr
[207]	Nanoparticles	Melting, numerical	The enhancement of melting of eicosane (PCM) filled in a heated bottom rectangular cavity by integrating various carbon nanotube (CNT) charges (fillers) into PCM was investigated	

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[208]	Nanoparticles	Melting, numerical	The enhancement of melting of water (PCM) in a bottom-heated rectangular cavity by adding copper nanoparticles to the PCM was examined	The addition of nanoparticles enhances the melting process of PCM by improving the thermal conductivity and decreasing the latent heat of melting The melt fraction and energy stored increase by increasing the nanoparticle's volume fraction
[209]	Nanoparticles	Melting, numerical	The melting enhancement of n-eicosane (PCM) used in a heat sink by integrating (inserting) different types of nanoparticles was studied. Two cases were studied: the case where the PCM is nano-enhanced by nanoparticles of Al ₂ O ₃ , ZnO, or CuO or Cu, as well as the case where the PCM is nano-reinforced by hybrid nanoparticles	It was found that the insertion of 4% Cu (case of adding only one type of nanoparticles) into the PCM decreases the operating temperature and the duration of the latent phase change (melting time) by 4.69% and 3.33%, respectively, compared to the case of pure PCM On the other hand, it was demonstrated that the addition of hybrid nanoparticles composed of both 1% Al ₂ O ₃ and 3% Cu to PCM resulted in a 5.77% decrease in operating temperature and a 31.11% increase in melting time; so, an improvement in the effective thermal effusivity of Ne-PCM of 10.85% compared to the case of adding a single type of nanoparticle (first case)
[210]	Nanoparticles	Melting, experimental	The enhancement of melting of 1-dodecyl (PCM) filled in a bottom-heated vertical cylindrical cavity by the inclusion of multi-walled carbon nanotubes (CNTs) with different fillers (loadings) was studied	It was found that the inclusion of CNTs in PCM improves its thermal conductivity. Additionally, higher fillers of CNTs were found to be beneficial. Indeed, this considerably increases the viscosity of the PCM, which degrades natural convection, which in turn slows down the melting process
[211]	Nanoparticles	Melting, experimental	The melting enhancement of 1-dodecyl (PCM) filled in a spherical chamber by dispersion of self-synthesized graphite nanosheets (GNs) with different fillers in the PCM was studied	It was found that the presence of nanofillers in PCM increases its thermal conductivity and viscosity; this significantly degrades natural convection, causing the melting process to slow down
[212]	Nanoparticles	Melting, numerical	The study of melting enhancement of paraffin enhanced by aluminum oxide (Al ₂ O ₃) as a Ne-PCM filled in a square enclosure using different configurations of combinations of a couple (heat source-heat sink) placed on its vertical side walls was carried out	It was found that the liquid fraction is higher when sources and sinks are located alternately on two vertical side walls. In addition, the liquid fraction turns out to be lower when the sources are located above the sinks on the vertical side walls On the other hand, it has been shown that the highest melting rate is found at the nanoparticle's volumetric concentration of 2 vol%

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[213]	Nanoparticles	Melting, numerical	The melting enhancement of n-octadecane (PCM) filled in a rectangular cavity by the insertion of CuO nanoparticles with different volume fractions into the PCM was examined	Insertion of nanoparticles into PCM has been shown to improve heat transfer and melting rate by increasing its thermal conductivity. In addition, the melting time was found to be reduced with increase in nanoparticle volume fraction. On the other hand, the addition of nanoparticles has been shown to increase “nano-enhanced PCM” fluid viscosity, which decreases the melting rate and heat transfer rate
[214]	Nanoparticles	Melting, numerical	Fatty acid-based PCM dispersed with various percentages of graphene nanoplatelets (GNPs) ranges from 0.1 % to 0.5% inside a spherical capsule	The PCM melting time reduces with the increase of the graphene nanoplatelets (GNPs) charging; it decreased by 31.2% and 28.1% by using 0.5% of graphene and for the PCM initial temperature of -10°C and 2°C , respectively
[215]	Nanoparticles	Melting, numerical	The enhancement of paraffin (PCM) melting filled in a horizontally heated (from the bottom wall) or vertically heated (from the side wall) square cavity by the insertion of alumina nanoparticles was investigated	It was found that increasing the volume concentration of Al_2O_3 decreases the melting rate. On the other hand, it was found that for pure paraffin, the “melting rate” and “stored energy” are, respectively, 25.4% and 21.2% higher, when the cavity is heated vertically than when the cavity is heated horizontally, while for paraffin containing 2% Al_2O_3 , the “melting rate” and “stored energy” are, respectively, 22.9% and 19.3% higher when the cavity is heated vertically than when the cavity is heated horizontally
[216]	Nanoparticles	Melting, numerical	The process of melting paraffin wax with 1 vol% silicon carbide (SiC) nanoparticles inside a rectangular cavity was studied	The use of SiC nanoparticles has been shown to improve the liquid fraction f_l of PCM and the melting time t_m by 33% as well as the latent heat storage capacity by 50% compared to pure PCM
[217]	Nanoparticles	Solidification, numerical	Improving water solidification (PCM) inside a vertical corrugated (wavy) wall cavity by inserting Cu nanoparticles with different volume fractions was investigated. The whole (cavity and Cu-water nano-enhanced PCM) is subjected to different values of the Grashof number and different values of surface waviness	Inserting nanoparticles into water (PCM) was found to reduce solidification time. On the other hand, increasing surface waviness was shown to accelerate the solidification process by 25.9% with 0.5 surface waviness for all Grashof number values tested

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[218]	Nanoparticles	Solidification, numerical	The solidification enhancement of n-hexadecane (PCM) filled in a vertical cavity by inserting Cu nanoparticles into the PCM was examined	It was concluded that the dispersion of nanoparticles significantly increases the heat transfer rate. On the other hand, it was found that the solid fraction is greater when the wall temperature is lower and the volume fraction of nanoparticles is higher. Furthermore, increasing the volume fraction of nanoparticles leads to an increase in the heat transfer rate.
[219]	Nanoparticles	Solidification, numerical	A colloidal model of water (PCM) solidification improvement reinforced by the insertion of copper nanoparticles of different sizes was developed	It was found that for a nanoparticle size of 5 nm, the solid–liquid interface was almost flat during the solidification process. Similarly, it was found that for a particle size of 2 nm, the solid–liquid interface developed from a stable planar morphology to an unstable dendritic morphology. It was concluded that decreasing the size of the nanoparticles thins the frozen colloidal layer.
[220]	Nanoparticles	Solidification, numerical	The solidification enhancement of dodecanol (PCM) by the insertion of graphene nanoplatelets was examined	It was found that increasing the charging (load) of the nanoparticles improves solidification; it is strongly accelerated by 34% for 1 mass%.
[221]	Nanoparticles	Solidification, numerical	Enhancing solidification of water (PCM) filled in an LHS storage unit by insertion of copper (Cu) nanoparticles with different values of nanoparticle volume fractions. The two shapes of the storage unit, the square shape and the isosceles trapezoidal enclosure shape, were tested and compared. The assembly (ne-PCM and storage unit) was subjected to different inclination angles and different values of the Grashof number	It was found that the “Ne-PCM” heat transfer is greatly improved by using a trapezoidal shaped storage unit compared to the square-shaped one. On the other hand, it was found that the solidification time is reduced by 11%, 11.5%, and 7% by increasing the inclination angle to 5.42°, 7.69°, and the volume fraction to 20%, respectively. Furthermore, it was shown that heat transfer is dominated by conduction for low Grashof values ($\leq 10^5$); while it is dominated by convection for higher Grashof values ($> 10^5$).
[222]	Nanoparticles	Melting & Solidification, experimental	Titanium dioxide (Titania) TiO_2 nanoparticles were dispersed in paraffin (PCM) with and without sodium stearoyl lactylate (SSL) as a surfactant. The objective was to examine the improvement in melting and solidification performance of the nanocomposite paraffin/ TiO_2 with and without SSL insertion	The maximum latent heat for the paraffin/ TiO_2 without SSL was shown to be produced at a concentration of 1 mass% of TiO_2 ; while with SSL it is produced at a concentration of 3 mass% of TiO_2 . Furthermore, thermal conductivity was found to be higher when using SSL; it increases by 47.85% at the concentration of 3 mass%. It was concluded that the presence of SSL improves the thermal stability and distribution of nanoparticles in the paraffin/ TiO_2 nanocomposite.

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[223]	Nanoparticles	Melting & Solidification, numerical	Graphene nanoplatelets (GNPs) were dispersed in paraffin (PCM) and the resulting paraffin-GN nanocomposite was filled inside a horizontal cylinder used as an LHS storage system. The objective was to examine the improvement in melting and solidification performance of the paraffin-GN nanocomposite	Adding GN to PCM has been shown to improve the melting and solidification process. Furthermore, it was found that the solidification rate increases with increase in nanoparticle charging. Indeed, the use of fractions 1 and 3 mass% of GN reduces the solidification time by 38% and 50%, respectively. On the other hand, the melting time has been shown to increase with increase in GN dispersion, except 3 mass%
[224]	Fin & nanoparticles	Melting, numerical	A study on improving the melting performance of a “proposed” shell-and-tube heat storage unit using a combination of optimal fin design and nanoparticle insertion was carried out	It was shown that long fins should be mounted at the bottom and short fins at the top. On the other hand, it was found that the melting rate increases with the increase in the volume fraction of the inserted nanoparticles. It was concluded that the “proposed” optimal design could improve the melting rate by 20%.
[225]	Fin & nanoparticles	Melting, numerical	The effects of including CuO nanoparticles in pure PCM and using “downward” and “upward” stair-shaped fins with different staircase ratios ($0.67 < b/c < 4.0$) on melting improvement were examined	The addition of CuO nanoparticles, with a volume fraction of 1.5% and for a b/c ratio = 0.67, has been shown to increase the “energy storage capacity” by approximately 9.1% compared to pure PCM. Likewise, it was found that the use of “downward” stair-shaped fins with a “ b/c ” ratio of 4 and the inclusion in the PCM of a volume fraction of 1.5% nanoparticles, make it possible to achieve “maximum energy storage capacity” of 474.1 kJ. “Downward” stair-shaped fins have also been shown to be more effective than “upward” stair-shaped fins.
[226]	Fin & nanoparticles	Solidification, numerical	Examining the enhancement of solidification of a PCM, filled in a triplex tube latent heat storage (LHS) system, by the use of V-shaped fins and the inclusion of SWCNT nanoparticles in the PCM has been achieved	High fin length has been shown to improve solidification. On the other hand, the inclusion of “SWCNT” nanoparticles with volume fractions of 2.5 and 5 vol% in PCM was shown to accelerate the solidification process by 1.16 and 1.49 times, respectively, compared to pure PCM. It was concluded that the use of V-shaped fins is more suitable than the dispersion (inclusion) of nanoparticles to improve the solidification process.

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[227]	Fin & nanoparticles	Solidification, numerical	Enhancing the solidification of filled water (PCM) in an LHS storage system by the use of branched tree-shaped fins and the inclusion (dispersion) of hybrid nanoparticles (TiO ₂ -Cu) to the PCM was studied. The assembly (LHS system and the hybrid nanocomposite (water/TiO ₂ -Cu)) are subjected to thermal radiation	It was found that the dispersion of hybrid nanoparticles involves a higher solidification rate. In addition, the higher form factor of nanoparticles has been shown to increase the solid fraction. On the other hand, it has been shown that by increasing the thermal radiation parameter, the average temperature and overall energy of the system decrease; while the solid fraction increases and the solidification process accelerates It has also been reported that a higher bifurcation angle (of branched tree-shaped fins) leads to a faster solidification process
[228]	Fin & nanoparticles	Solidification, numerical	The separate effects of (i) the use of fins of variable thickness at the LHS configuration and (ii) the inclusion (dispersion) of hybrid nanoparticles of (Al ₂ O ₃ -Go), of different volume fractions, at the PCM, on improving solidification processes were examined	It was found that increasing the length and number of fins improves the solidification rate. On the other hand, the use of Al ₂ O ₃ -Go hybrid nanoparticles at volume fractions of 2.5% and 5% was shown to accelerate the solidification process by 1.744 and 2.698 times, respectively, compared to pure PCM It was concluded that it is better to use an optimized fin configuration to improve LHS performance rather than dispersing hybrid nanoparticles in PCM
[229]	fin, foam, & nanoparticles	Solidification, numerical	Improving solidification in an LHS storage system by adding fins (having different configuration shapes) to the storage unit, (ii) incorporating foams into the PCM and (iii) dispersing nanoparticles into the PCM has been studied	It was found that the solidification time of PCM was improved by 42.42% and 39.39% in the type “3” fin configuration and type “5” fin configuration, respectively, compared to the fin configuration of type “4” Similarly, solidification time is reduced by 73.68% through the incorporation of nanoparticles and metal foam into PCM
[230]	External field	Melting, numerical	Melting improvement of a PCM filled in a square enclosure using two nonuniform magnetic fields was examined	It was shown that the evolution of the melting front depends entirely on the intensity ratio of two nonuniform magnetic fields In addition, increasing the Hartmann number has been shown to reduce the melting rate However, increasing the magnetic number has been shown to slightly increase the PCM melt fraction

Table 6 (continued)

References	Method	Process/type of study	Description and parametric details	Main results
[231]	External field	Melting, numerical and experimental	The enhancement of the melting of dodecanol (PCM), filled into a laterally heated cylinder and dispersed by magnetic particles, by the action of a rotating magnetic field was investigated	It was shown for a rotation speed of 20 rpm, a particle fraction of 1.0 mass% and a heating temperature of 35 °C, the total melting time is decreased by 22.9%
[232]	External field	Melting, experimental	The enhancement of PCM paraffin melting, filled in a latent heat storage (LHS) unit, using ultrasound was investigated	It was shown that by using ultrasound, the melting time was reduced by 60.69% and the heat transfer coefficient was increased by 25.97%
[233]	External field	Melting, numerical and experimental	The melting enhancement of a Ne-PCM composed of (n-octadecane and graphene) filled in a latent heat storage (LHS) system by the action of an electric field was studied	Applying a voltage of 25.0 kV has been shown to increase the energy absorbed (stored) by 153.5%. Furthermore, it was reported that when + 10.0 kV and -10.0 kV are applied, the stored energy increases by 111.2% and 124.8%, respectively
[234]	External field	Solidification, experimental	The enhancement of solidification of paraffin (PCM), enhanced by graphene nanoparticles, by the action of an electric field of 110 V and a magnetic field (magnetic induction of 0.1 T) was examined	Electromagnetic fields have been shown to improve the dispersion of graphene nanoparticles in PCM and increase the stability of the solidification cycle
[235]	External field	Solidification, numerical	The effect of the presence of a magnetic field on enhancing the solidification of a Ne-PCM, filled in a latent heat storage system (LHS), was studied	It was found that the solidification rate of Ne-PCM was improved with the increase in Lorentz force
[236]	External field	Solidification, numerical and experimental	The improvement of the solidification of a PCM melt under the action of a vibrating surface was studied	Vibration has been shown to increase the rate of solidification and accelerate the rate of nucleation In addition, it was found that increasing the vibration intensity increases the heat transfer coefficient
[237]	External field	Melting & solidification, numerical	The enhancement of the solidification of gallium (PCM), filled in an isosceles triangular cavity, by the action of a magnetic field uniformly applied in various directions and with different magnitudes (characterized by different values of the Hartmann number, Ha)	The total melting time was shown to increase by 32.5% and 5.5% using horizontal and vertical magnetic fields, respectively However, the magnitude and direction of the magnetic field does not affect the solidification process

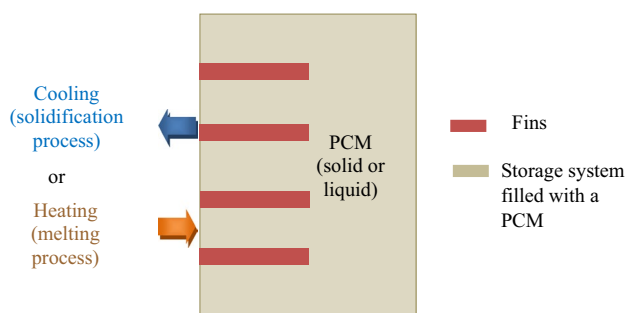


Fig. 26 Schematic of a latent heat storage system, filled with a PCM, which uses the fin integration enhancement technique. Generally, the plate-shaped fins are installed (embedded) horizontally on the PCM domain

Indeed, the morphology and material constituting the foam have a significant effect on the melting and solidification processes of the product (PCM doped with foam) [184]. In this context, it is noted that the integration of copper foam is more effective than aluminum foam for the melting and solidification processes of the compound (foam-doped PCM). In addition, these two processes are more efficient if the metal foams are characterized by low porosity and/or a larger pore size [185].

Enhancement by inclusion (dispersion) of highly conductive nanoparticles

Principle and discussion

Heat transfer from a PCM can be effectively enhanced by fabricating a composite, consisting of a dispersion of nanoparticles, with high thermal conductivity, and the base PCM

to form what is called a nano-enhanced-PCM (Ne-PCM), as schematized in Fig. 28.

The effect of nanoparticles on improving the thermal (heat transfer) performance of Ne-PCM depends on their nature (type of material composing these nanoparticles) and their volume fraction (i.e., porosity). Generally, copper (Cu), alumina (Al_2O_3), copper oxide (CuO), titanium dioxide (TiO_2) and graphene are the most widely used nanoparticles. In this context, a review of the literature, consulted as part of this study (e.g., see Table 6), showed that the melting phase change process can be improved by increasing the volume fraction of the nanoparticles. It also showed that the inclusion of Cu and Al_2O_3 nanoparticles improved the melting rate and therefore the heat stored (absorbed) in the compound PCM (Ne-PCM). However, it should be noted that this trend diminishes over time in the case of Al_2O_3 nanoparticles. While, in the case of the inclusion of copper nanoparticles, the enhancement presents irregular and intense variations during the first seconds of the process; then stabilizes as mentioned, for example, in the results of Nitsas et al. [186].

On the other hand, literature analysis also revealed that the maximum and minimum latent heat released during the solidification process is obtained when including dispersions of TiO_2 and Al_2O_3 nanocomposites, respectively. Furthermore, it showed that the highest and lowest improvement, in the thermal conductivity of the composite PCM, was attributed to the inclusion of graphene and TiO_2 nanoparticles in the PCM, respectively. An example of a study describing these trends is that carried out by Aziz et al. [187].

Figure 29 shows photographs of paraffin (pure PCM) and various nanoparticles, mainly aluminum oxide, titanium dioxide, and copper oxide nanoparticles. These products are our samples which will be used in future experimental work and which will deal with the technique of enhancing

Fig. 27 Schematic of a latent heat storage system that uses a combination of metal foam with PCM as an enhancement technique

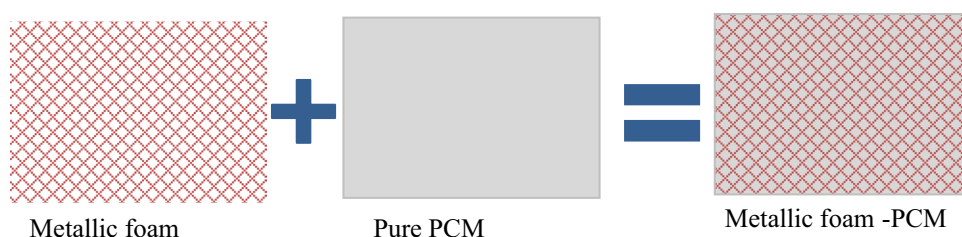


Fig. 28 Schematic of a latent heat storage system, filled with a PCM, which uses an insertion of highly conductive dispersed “nanoparticles” as an enhancement technique

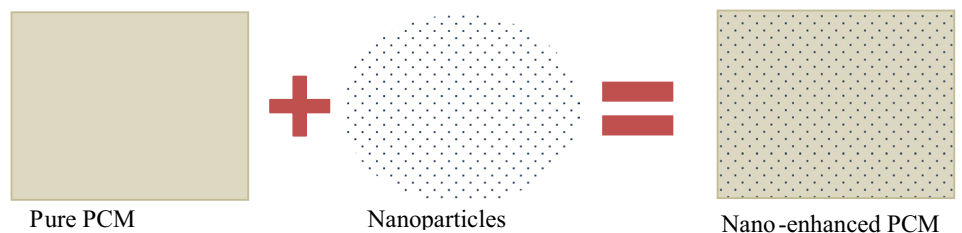


Fig. 29 Photographs of pure paraffin and various nanoparticles that will be used in our future experimental work aimed at improving performance during PCM phase transitions by including nanoparticle dispersions



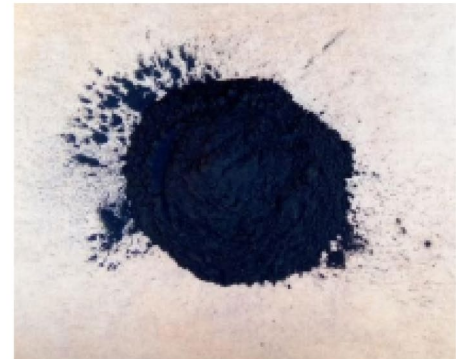
Paraffin PCM



Aluminum oxide nanoparticles



Titanium dioxide nanoparticles



Copper oxide nanoparticles

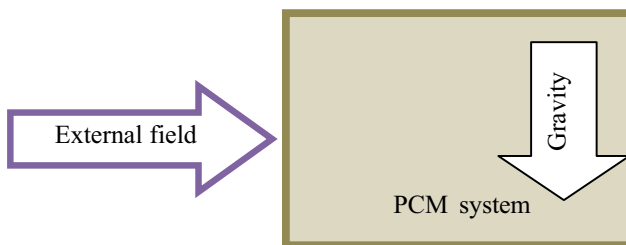


Fig. 30 Schematic of a latent heat storage system filled with a PCM and using the external field as an enhancement technique [188]

heat transfer during phase transitions using the inclusion of nanoparticle dispersions.

Enhancement by applying external force fields to the PCM system

Principle and discussion

The application of external fields, including electric field, magnetic field, ultrasound, or mechanical vibration, allows dynamic manipulation of the performance of the PCM system (e.g., LHS unit, see Fig. 30) by adjusting the additional energy input, caused by the applied external agent, which leads to an improvement in the melting and/or solidification processes of PCM in the system. In this context, the analysis

of the literature consulted as part of this study (e.g., see Table 6) showed that: the application of ultrasonic waves to the PCM system during the solidification process increases the probability of nucleation and thus reduces the induction time, which favors the process. Ultrasonics can also reduce the cooling energy required for primary nucleation and stabilize the energy stored (released) during solidification. On the other hand, the analysis revealed that applying a magnetic field to the PCM system can reduce the solidification and melting time of PCM. However, applying a high-frequency electric field to the PCM system during the PCM solidification process results in a large number of crystals; this also increases cycle capacity and ensures stable stored (released) energy.

Finally, it should be emphasized that external fields applied to the PCM system can also improve heat transfer during a phase change when this transition is dominated by the natural convection mode or by boiling, as, for example, reported by Wu et al. [188].

Conclusions

Renewable energy studies are important for the sustainable development of thermal energy storage. Among the latter, we distinguish latent heat storage systems using PCMs. The latter can absorb or release heat during their phase

transition from solid–liquid or liquid–solid, respectively. Through this review of the literature, we reached the following conclusions:

- PCMs are substances capable of absorbing or releasing a large amount of energy during their phase change. They are classified according to their nature and properties into three categories: organic, inorganic, and eutectic. Each category is designed for specific practical applications.
- PCMs are used in a variety of applications, including thermal storage of latent heat. On the other hand, they can be used for example: In the field of buildings to regulate the temperature of the building by mixing the PCM with the building material or by encapsulating it in the bricks. HVAC systems can be integrated with these systems to reduce energy demand and improve the efficiency of these systems. In electronics, PCMs can stabilize the temperature and improve the thermal control of electronic equipment by absorbing the intense heat generated during their operation. In the textile industry, PCMs can maintain a uniform temperature for the human body and improve thermal comfort if microencapsulated in the fabric. In the medical and biomedical fields, they facilitate the storage and transport of medical and biological products by maintaining a stable and adequate temperature. In agriculture and food, to maintain greenhouse temperatures, transportation and storage of products at appropriate levels. Battery temperature regulation in automotive manufacturing to improve thermal comfort of vehicles. Desalination consumes less energy and has higher thermal efficiency.
- The melting process of a PCM (storage phase) is initially characterized by heat transfer dominated by conduction. Over time, the temperature in the system increases and melting progresses, heat transfer becomes dominated by convection which occurs in the molten PCM and promotes heat transfer therein. In contrast, the solidification process of a PCM follows the opposite behavior; in fact, the heat transfer takes place mainly by conduction due to the high thermal conductivity of the resulting solid PCM (formed during solidification) compared to the liquid PCM.
- The heat transfer performance during the phase change of a PCM is affected by several parameters such as the geometry and boundary conditions imposed on the PCM system, its orientation and the properties of the material (PCM).
- The use of a PCM characterized by high thermal conductivity, latent heat, or viscosity and a low melting point (temperature) accelerates both its melting and solidification by reducing their duration.
- The melting process can be accelerated, leading to an increase in stored energy, both by imposing a high value on the wall from which the process begins and by using a sufficiently heated PCM melt (at an initial temperature sufficiently high). On the other hand, the solidification process decelerates with the increase in the temperature imposed on the wall from which solidification begins and with the increase in the initial temperature of the PCM melt.
- The orientation of the PCM system (e.g., LHS unit) is crucial for the melting and solidification processes. We found that increasing the inclination angle of the PCM system decreases the melting rate; conversely, it accelerates the solidification process.
- PCM melting and solidification rates and therefore stored energy can be significantly improved: by (1) the incorporation (inclusion) of high thermal conductivity additives such as metal foams or nanoparticle dispersions in pure PCM, and/or (2) the installation of fins to the PCM system, and/or (3) the application of external fields to the PCM system considered (such as the LHS storage unit for example). In these contexts, the following points should be emphasized:
 - (a) Adjusting the number, height, length, and orientation of fins installed in the PCM system can significantly improve its storage efficiency.
 - (b) Incorporation of low porosity, large pore metal foam into pure PCM reduces the time required for the phase change process, increasing its storage efficiency.
 - (c) The inclusion of nanoparticle dispersions with high thermal conductivity into pure PCM, leading to nano-enhanced PCM with increased thermal conductivity and latent heat of phase change. As a result, the melting and/or solidification processes (of this nano-enhanced PCM (Ne-PCM)) lead to increased storage efficiency.
 - (d) The application of external force fields, particularly the magnetic field, can be used effectively in enhancing the phase change process; this is due to the Lorentz force [238] which serves to stabilize the convection in the liquid phase [102] which results in a reduction in the time of phase change processes which in turn leads to an improvement in the thermal energy storage efficiency of the PCM. On the other hand, the application of an electric field has beneficial effects on the rate of heat transfer in the PCM during its phase change, which results in a clear improvement in its thermal energy storage capacity. Finally, the application of ultrasonic waves can improve the solidification process by increasing the nucleation possibilities and reducing the induction time which is beneficial to the storage efficiency of PCM.

Future research and perspectives

This work, like others before it, is limited by the low range of PCMs examined, as well as the lack of experimental investigations and validations, particularly concerning PCM solidification and integration in current systems. Future studies should expand the scope of PCMs examined and validate theoretical models through testing while keeping these limits in mind. To develop a more thorough understanding of PCM performance under various settings, future studies should examine a wider range of PCMs. Additionally, investigating the effects of boundary conditions and geometric layouts will improve the suitability of PCMs in actual thermal energy systems. Innovative design techniques for raising the effectiveness of thermal energy storage technologies may result from these studies.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships which might appear to influence the work reported in this article.

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